

Definitions and Facts in Plane Geometry

Version 3.2

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Topics

This note includes the following topics.

- triangles
 - centroid
 - circumcircle and circumcentre
 - orthocentre
 - nine-point circle and nine-point centre
 - incircle and incentre
 - excircles and excentres
 - mixtilinear incircles
 - pedal triangles
 - isogonal conjugate
 - Feuerbach point
 - symmedian point
 - humpty points
 - dumpty points
 - Apollonius circles
 - isodynamic points
 - Brocard points
 - Napoleon triangles
 - Fermat points
 - Spieker centre
 - Schiffler point
 - spiral similarity
- quadrilaterals
 - Miquel point
 - trapezoid
 - parallelogram
 - kite
 - rhombus
 - rectangle
 - square
 - cyclic quadrilateral
 - circumscribed quadrilateral
 - bicentric quadrilateral
- circles

Notations

Here gives a list of notations used in this note, most of which are not standard notations (meaning that one should define the symbols when they are used).

- (1) to (5): the number indicates how well-known the fact is, where 5 means the most well-known and 1 means the least well-known
- (1) to (5): the number indicates how useful the fact is, where 5 means the most useful and 1 means the least useful
- (Def): this means an item which is a definition
- $\sim$: this indicates there are other similar expressions or results due to symmetry
- $\ell_1 \cap \ell_2$: the intersection point of the lines ℓ_1 and ℓ_2
- $\ell_1 \cap \ell_2 \cap \dots \cap \ell_n$: the lines (or circles) $\ell_1, \ell_2, \dots, \ell_n$ are concurrent
- $\ell(P)$: an arbitrary line passing through the point P
- M_{PQ} : the midpoint of P and Q
- $\text{Rf}(P, Q)$: the reflection of the point P in the point Q
- $\text{Rf}(P, \ell)$: the reflection of the point P in the line ℓ
- $\text{Rf}(\ell_1, \ell_2)$: the reflection of the line ℓ_1 in the line ℓ_2
- $\perp(P, \ell)$: the perpendicular line through the point P to the line ℓ
- $\perp P \cap \ell$: the foot of perpendicular from the point P to the line ℓ
- $\parallel(P, \ell)$: the parallel line through the point P to the line ℓ
- $d(P, \ell)$: the distance from the point P to the line ℓ
- $[P_1 P_2 \dots P_n]$: the area of the polygon $P_1 P_2 \dots P_n$
- $(P_1 P_2 \dots P_n)$: the circle that passes through the points $P_1, P_2, \dots, P_n$
- $\odot O_1$: an arbitrary circle with centre O_1
- $\odot(P, d)$: the circle with centre P and radius d
- $\tan(P, \Gamma)$: the tangent line at the point P (lying on Γ) to the circle Γ
- $\tilde{P}_\Gamma$: the power of the point P with respect to(w.r.t.) the circle Γ
- a, b, c : the side lengths BC, CA, AB of $\triangle ABC$
- a, b, c, d : the side lengths AB, BC, CD, DA of a quadrilateral $ABCD$
- R : the circumradius of $\triangle ABC$ or a cyclic quadrilateral $ABCD$

- r : the inradius of $\triangle ABC$ or a circumscribed quadrilateral $ABCD$
- s : the semiperimeter of $\triangle ABC$ or a quadrilateral $ABCD$
- r_A, r_B, r_C : the radii of the A, B, C -excircles of $\triangle ABC$
- M_A, M_B, M_C : the midpoints of the sides BC, CA, AB of $\triangle ABC$
- G : the centroid of $\triangle ABC$
- $\odot O$: the circumcircle of $\triangle ABC$ or a cyclic quadrilateral $ABCD$
- O : the circumcentre of $\triangle ABC$ or a cyclic quadrilateral $ABCD$
- H_A, H_B, H_C : the feet of altitudes from vertices A, B, C of $\triangle ABC$
- H : the orthocentre of $\triangle ABC$
- $\odot N$: the nine-point circle of $\triangle ABC$
- N : the nine-point centre of $\triangle ABC$
- I_A, I_B, I_C : the feet of the internal angle bisectors from vertices A, B, C of $\triangle ABC$
- $\odot I$: the incircle of $\triangle ABC$ or a circumscribed quadrilateral $ABCD$
- I : the incentre of $\triangle ABC$ or a circumscribed quadrilateral $ABCD$
- T_A, T_B, T_C : the contact points of the incircle and the sides BC, CA, AB of $\triangle ABC$
- Ge : the Gergonne point of $\triangle ABC$
- Sd_A, Sd_B, Sd_C : the A, B, C -Sharky-Devil points of $\triangle ABC$
- $\odot E_A, \odot E_B, \odot E_C$: the A, B, C -excircles of $\triangle ABC$
- E_A, E_B, E_C : the A, B, C -excentres of $\triangle ABC$
- U_A, U_B, U_C : the contact points of the A, B, C -excircles and the sides BC, CA, AB of $\triangle ABC$
- Na : the Nagel point of $\triangle ABC$
- $\Gamma_A, \Gamma_B, \Gamma_C$: the A, B, C -mixtilinear incircles of $\triangle ABC$
- V_A, V_B, V_C : the contact points of the A, B, C -mixtilinear incircles and the circumcircle of $\triangle ABC$
- $\text{Iso}(P)$: the isogonal conjugate of the point P w.r.t. $\triangle ABC$
- Fu : the Feuerbach point of $\triangle ABC$
- K : the symmedian point of $\triangle ABC$
- Hu_A, Hu_B, Hu_C : the A, B, C -humpty points of $\triangle ABC$

- Du_A, Du_B, Du_C : the A, B, C -dumpty points of $\triangle ABC$
- $\omega_A, \omega_B, \omega_C$: the A, B, C -Apollonius circles of $\triangle ABC$
- S_1, S_2 : the 1st and 2nd isodynamic points of $\triangle ABC$
- Br_1, Br_2 : the 1st and 2nd Brocard points of $\triangle ABC$
- ω : the Brocard angle of $\triangle ABC$
- F_1, F_2 : the 1st and 2nd Fermat points of $\triangle ABC$
- Sp : the Spieker centre of $\triangle ABC$
- Sc : the Schiffler point of $\triangle ABC$
- Mq : the Miquel point of a quadrilateral $ABCD$
- $\text{Inv}(P, \Gamma)$: the inversion of the point P w.r.t. the circle Γ
- $P_{\triangle XYZ}$: the corresponding special point P of $\triangle XYZ$ (instead of the notation defined for $\triangle ABC$)

Definitions and Facts

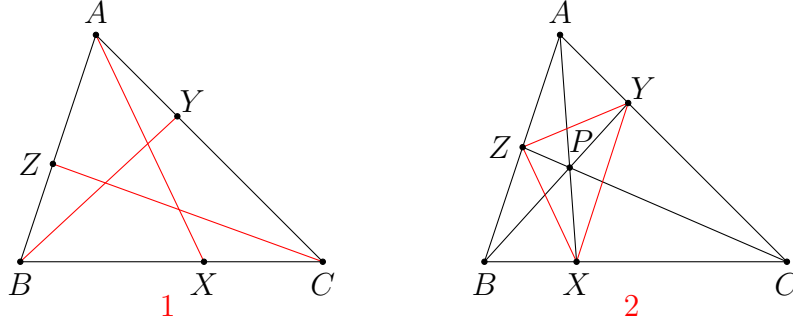
For most facts which we state for acute triangles, there are analogous results for the general cases. The acute conditions are added for convenience in many cases.

- (5)(5)(Def) In $\triangle ABC$, points X, Y, Z lie on the sides BC, CA, AB respectively. Then each of the segments AX, BY, CZ is called a *cevian*(塞瓦線段) of $\triangle ABC$.

(5)(5)(Def) In particular, AM_A, BM_B, CM_C are called the *medians*(中線) of $\triangle ABC$;

(5)(5)(Def) AH_A, BH_B, CH_C are called the *altitudes*(高) of $\triangle ABC$;

(5)(5)(Def) AI_A, BI_B, CI_C are called the *internal angle bisectors*(內角平分線) of $\triangle ABC$.



- (5)(5)(Def) In $\triangle ABC$, P is an interior point. Let $X = AP \cap BC$, $Y = BP \cap CA$, $Z = CP \cap AB$. Then $\triangle XYZ$ is called the *cevian triangle*(塞瓦三角形) of P w.r.t. $\triangle ABC$.

(5)(4)(Def) In particular, $\triangle M_A M_B M_C$ is the cevian triangle of G , called the *median triangle*(中點三角形) of $\triangle ABC$;

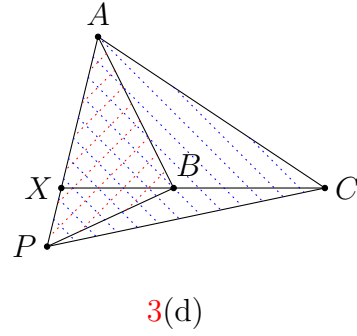
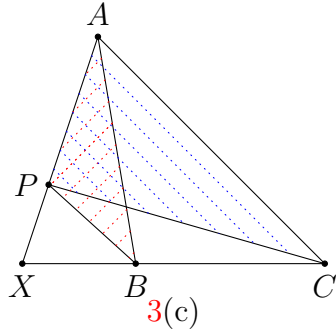
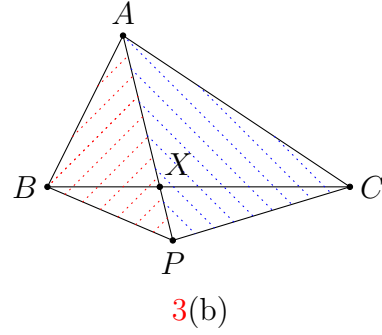
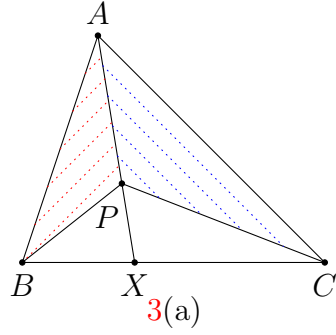
(5)(5)(Def) $\triangle H_A H_B H_C$ is the cevian triangle of H , called the *orthic triangle*(垂足三角形) of $\triangle ABC$;

(5)(3)(Def) $\triangle I_A I_B I_C$ is the cevian triangle of I , called the *incentric triangle* of $\triangle ABC$;

(5)(5)(Def) $\triangle T_A T_B T_C$ is the cevian triangle of Ge , called the *intouch triangle* of $\triangle ABC$;

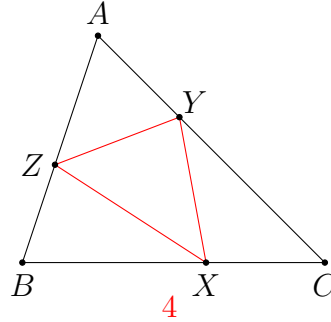
(5)(1)(Def) $\triangle U_A U_B U_C$ is the cevian triangle of Na , called the *extouch triangle* of $\triangle ABC$.
- (5)(4) In $\triangle ABC$, a point X lies on the line BC and a point P lies on the line AX . Then $BX : CX = [ABP] : [ACP]$.

(5)(4) In particular, $BX : CX = [ABX] : [ACX]$.



4. (2)(1) In $\triangle ABC$, points X, Y, Z lie on the sides BC, CA, AB respectively.

$$\text{Then } [XYZ] = \frac{AZ \cdot BX \cdot CY + AY \cdot BZ \cdot CX}{abc} \cdot [ABC].$$



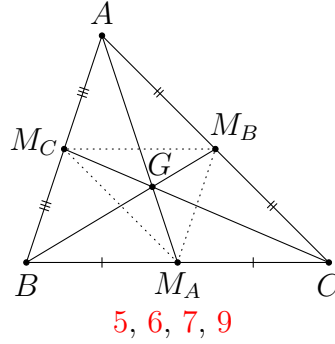
5. (5)(4)(Def) Let $\triangle ABC$ be a given triangle.

Then $AM_A \cap BM_B \cap CM_C = G$, called the *centroid* (重心) of $\triangle ABC$.

6. (5)(1) Let $\triangle ABC$ be a given triangle.

$$\text{Then } [AGM_B] = [AGM_C] = \frac{1}{6}[ABC] (\sim).$$

(5)(5) So $AG : GM_A = 2 : 1 (\sim)$.



7. (5)(3) (Apollonius' theorem(阿波羅尼斯定理)) Let $\triangle ABC$ be a given triangle.

Then $AM_A = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2} (\sim).$

8. (2)(1) Let $\triangle ABC$ be a given triangle and let P be any point.

Then $3(PA^2 + PB^2 + PC^2) = 9PG^2 + a^2 + b^2 + c^2.$

(3)(1) So $PA^2 + PB^2 + PC^2$ has the smallest value when $P = G.$

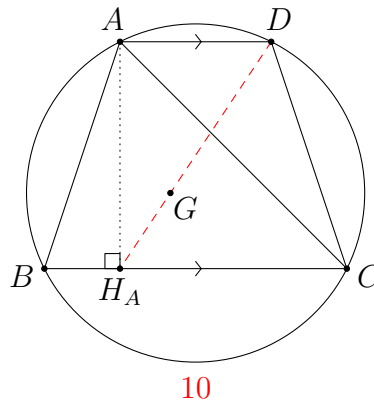
9. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $\triangle ABC$ and $\triangle M_A M_B M_C$ are homothetic with centre $G.$

(5)(1) So $G = G_{\triangle M_A M_B M_C}.$

10. (3)(1) Let $\triangle ABC$ be a given triangle. Let D be the point such that $ABCD$ is an isosceles trapezoid with $AD \parallel BC.$

Then H_A, G, D are collinear.



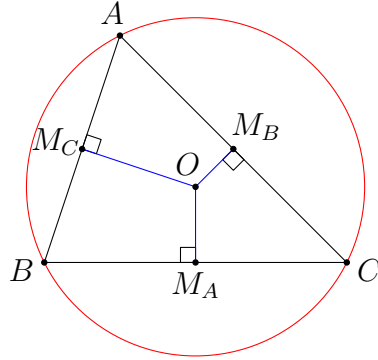
11. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then $\perp (M_A, BC) \cap \perp (M_B, CA) \cap \perp (M_C, AB) = O,$ called the *circumcentre*(外心) of $\triangle ABC.$

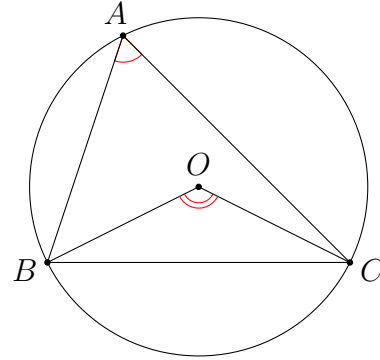
12. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then A, B, C lie on $\odot(O, R)$, called the *circumcircle*(外接圓) of $\triangle ABC$.

(5)(5) So $OA = OB = OC = R$.



11, 12, 15



13

13. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $\angle BOC = 2A$ (while $\angle BOC = 2\angle BAC$ always holds) ($\sim$) (see also fact 245).

14. (5)(5) Let $\triangle ABC$ be a given triangle.

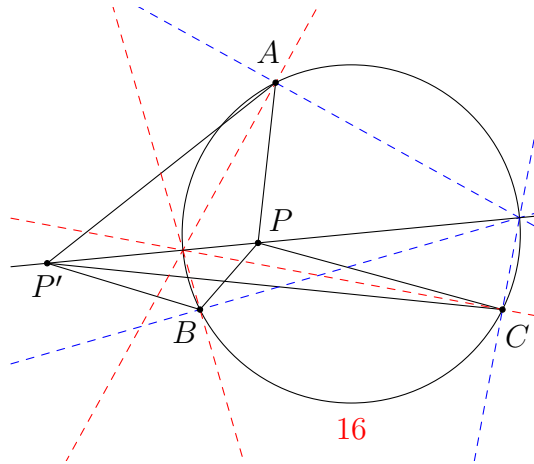
$$\text{Then } R = \frac{abc}{4[ABC]}.$$

15. (4)(1) (Carnot's theorem(卡諾定理)) Let $\triangle ABC$ be an acute triangle.

$$\text{Then } OM_A + OM_B + OM_C = R + r.$$

16. (1)(1) Let $\triangle ABC$ be a given triangle and let $P \neq A, B, C, O$ be any point. Let $P' = \text{Inv}(P, \odot O)$ and let ℓ_1, ℓ_2, ℓ_3 be the internal (or external) angle bisectors of $\angle PAP', \angle PBP', \angle PCP'$.

Then $\ell_1 \cap \ell_2 \cap \ell_3 \cap PP' \cap \odot O$.



16

17. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then $AH_A \cap BH_B \cap CH_C = H$, called the *orthocentre*(垂心) of $\triangle ABC$.

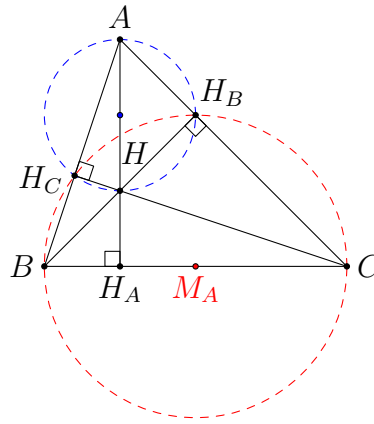
18. (5)(5) Let $\triangle ABC$ be a given triangle.

Then B, C, H_B, H_C are concyclic, lying on a circle with centre M_A ($\sim$).

(5)(3) So $M_A H_B = M_A H_C = \frac{a}{2}$ ($\sim$).

19. (5)(5) Let $\triangle ABC$ be a given triangle.

Then A, H_C, H, H_B are concyclic, lying on a circle with centre M_{AH} ($\sim$).



17, 18, 19, 20, 21, 22, 23

20. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $\angle BHC = 180^\circ - A$ (while $\angle BHC = -\angle BAC$ always holds) ($\sim$).

21. (5)(3) Let $\triangle ABC$ be a given triangle.

Then $AH_A = \frac{2[ABC]}{a}$ ($\sim$).

22. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $AH = 2R \cos A$ (while $AH = 2R |\cos A|$ always holds) ($\sim$).

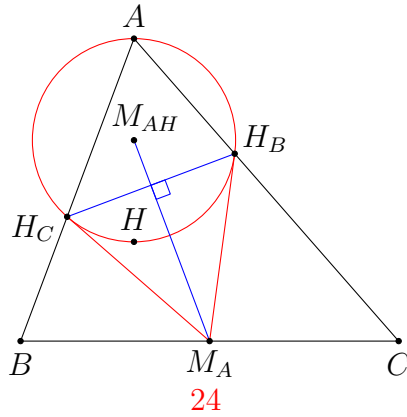
23. (5)(3) Let $\triangle ABC$ be a given triangle.

Then $A = H_{\triangle BCH}$ ($\sim$).

24. (4)(2) Let $\triangle ABC$ be a given triangle.

Then $M_A H_B$ and $M_A H_C$ are tangent to $(AH_C H H_B)$.

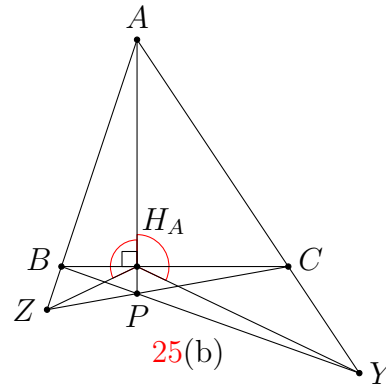
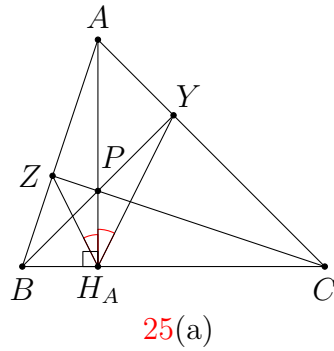
(4)(2) So $M_A M_{AH} \perp H_B H_C$.



25. (4)(1) (Blanchet's theorem) In acute $\triangle ABC$, a point P lies on the line AH_A . Let $Y = BP \cap AC$, $Z = CP \cap AB$.

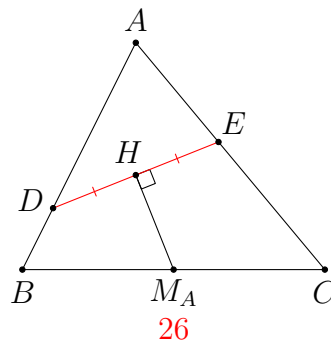
Then $\angle AH_A Y = \angle AH_A Z$.

(5)(3) In particular, $\angle AH_A H_B = \angle AH_A H_C$.



26. (2)(1) In $\triangle ABC$, let $D, E = \perp (H, HM_A) \cap AB, AC$.

Then $DH = EH$.



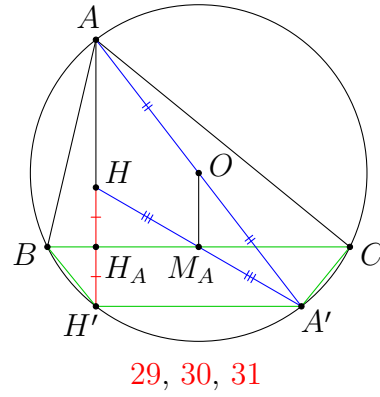
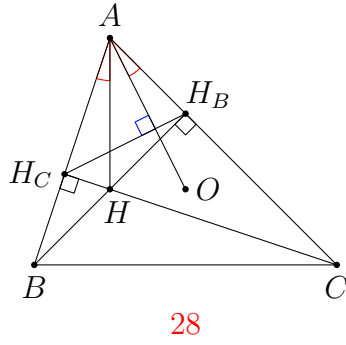
27. (4)(1) (Fagnano's problem(法格乃諾問題)) In acute $\triangle ABC$, points X, Y, Z lie on the sides BC, CA, AB respectively.

Then $\triangle XYZ$ has the smallest perimeter when $X, Y, Z = H_A, H_B, H_C$.

28. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $\angle CAO = \angle BAH$ and $\angle BAO = \angle CAH$ ($\sim$).

(5)(4) So $AO \perp H_B H_C$ ($\sim$) (see also fact 125).



29. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $\text{Rf}(H, M_A) = \text{Rf}(A, O)$ ($\sim$).

(5)(5) So $\text{Rf}(H, M_A)$ lies on $\odot O$ and $AH = 2OM_A$ ($\sim$).

30. (5)(5) Let $\triangle ABC$ be an acute triangle.

Then $\text{Rf}(H, BC)$ lies on $\odot O$ ($\sim$).

(4)(2) So $H_A B \cdot H_A C = H_A A \cdot H_A H$ ($\sim$).

31. (5)(4) In acute $\triangle ABC$, let $A' = \text{Rf}(H, M_A)$ and $H' = \text{Rf}(H, H_A)$.

Then $BCA'H'$ is an isosceles trapezoid with $BC \parallel H'A'$ ($\sim$).

32. (2)(1) In acute $\triangle ABC$, let $H' = \text{Rf}(H, H_A)$.

Then $M_{BH'}OM_BH_A$ is a parallelogram ($\sim$).

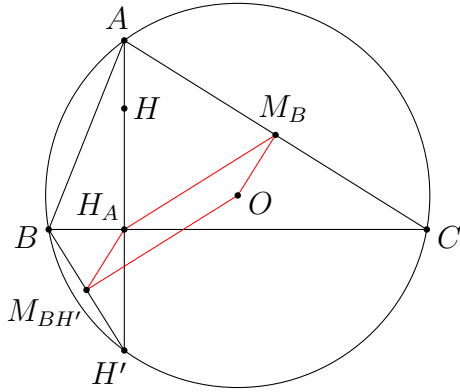
Limiting case: when $\angle C = 45^\circ$, $M_{BH'}, H_A, O, M_B$ are collinear ($\sim$).

(2)(1) In particular, $BH' \perp M_B H_A$ ($\sim$).

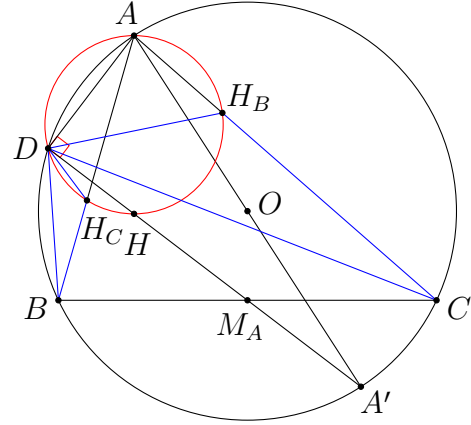
33. (5)(5) In acute $\triangle ABC$, let $A' = \text{Rf}(A, O)$ and $D = HM_A \cap \odot O$ (other than A').

Then D lies on $(AH_C H H_B)$ ($\sim$).

(5)(5) So $\angle ADH = 90^\circ$ ($\sim$).



32



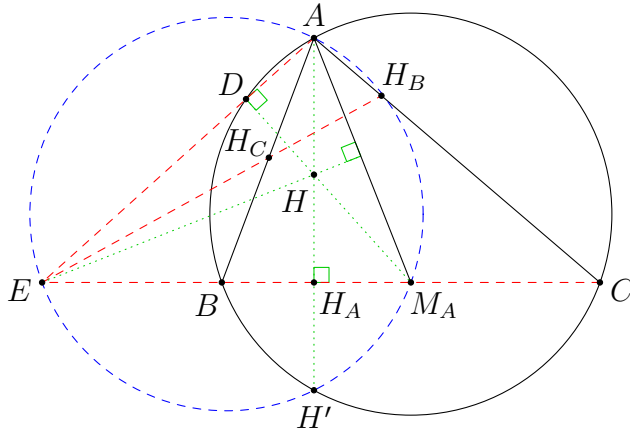
33, 34

34. (5)(4) In acute $\triangle ABC$, let $A' = \text{Rf}(A, O)$ and $D = HM_A \cap \odot O$ (other than A').
Then $\triangle DBH_C \sim \triangle DCH_B$ ($\sim$).

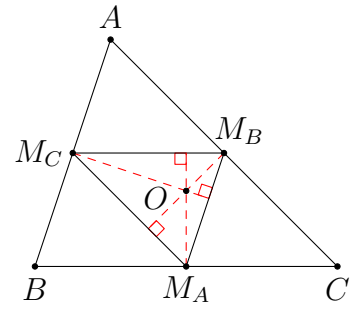
35. (3)(2) In acute $\triangle ABC$, let $D = \odot O \cap (AH_C HH_B)$.
Then $BC \cap H_B H_C \cap AD$ ($\sim$).

(3)(1) In particular, let E be the intersection point and let $H' = \text{Rf}(H, H_A)$.
Then A, E, H', M_A are concyclic and $H = H_{\triangle AEM_A}$ ($\sim$).

(3)(1) So $AM_A \perp EH$ ($\sim$).



35



36

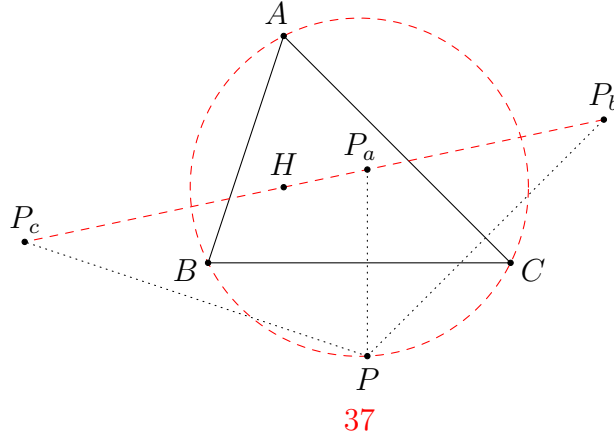
36. (5)(2) Let $\triangle ABC$ be a given triangle.
Then $O = H_{\triangle M_A M_B M_C}$.

37. (5)(2) In $\triangle ABC$, let P be any point. Let $P_a = \text{Rf}(P, BC)$, $P_b = \text{Rf}(P, CA)$, $P_c = \text{Rf}(P, AB)$.

Then P_a, P_b, P_c are collinear if and only if (iff) P lies on $\odot O$.

(5)(2)(Def) In particular, when P lies on $\odot O$, the line on which P_a, P_b, P_c lie is called the *Steiner line* (斯坦納線) of P w.r.t. $\triangle ABC$.

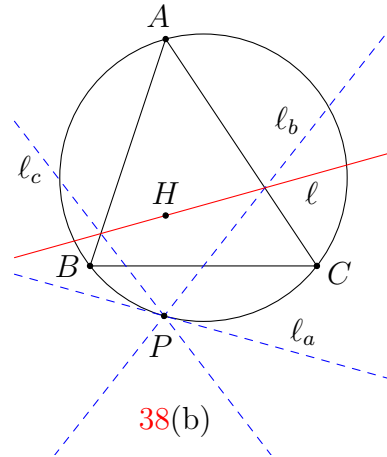
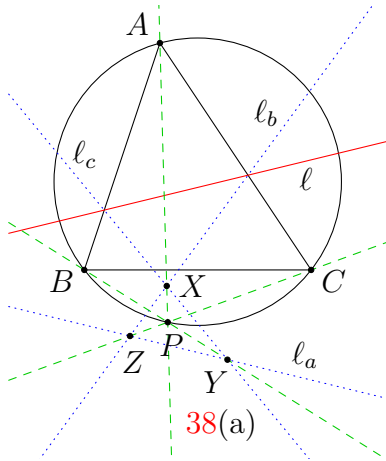
(5)(2) The Steiner line must pass through H (see also fact 122).



38. (1)(1) Let $\triangle ABC$ be a given triangle. Let ℓ be any line. Let $\ell_a = \text{Rf}(\ell, BC)$, $\ell_b = \text{Rf}(\ell, CA)$, $\ell_c = \text{Rf}(\ell, AB)$, $X = \ell_b \cap \ell_c$, $Y = \ell_c \cap \ell_a$, $Z = \ell_a \cap \ell_b$.

Then $AX \cap BY \cap CZ \cap \odot O$.

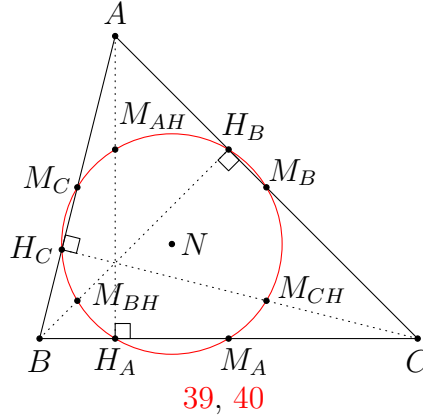
(4)(1)(Def) In particular, $\ell_a \cap \ell_b \cap \ell_c$ iff H lies on ℓ . The intersection point is called the *anti-Steiner point* (逆斯坦納點) of ℓ w.r.t. $\triangle ABC$.



39. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then $M_A, M_B, M_C, H_A, H_B, H_C, M_{AH}, M_{BH}, M_{CH}$ are concyclic, lying on the *nine-point circle* (九點圓/費爾巴哈圓) of $\triangle ABC$.

(5)(5)(Def) The centre of the nine-point circle is N , called the *nine-point centre* (九點圓圓心) of $\triangle ABC$.



40. (5)(4) Let $\triangle ABC$ be a given triangle.

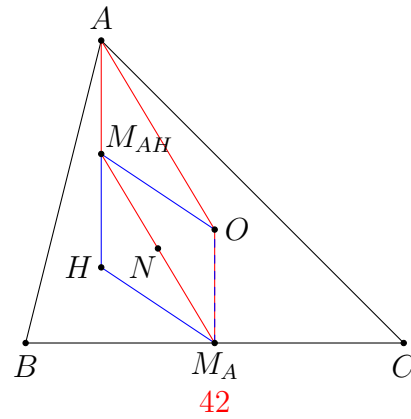
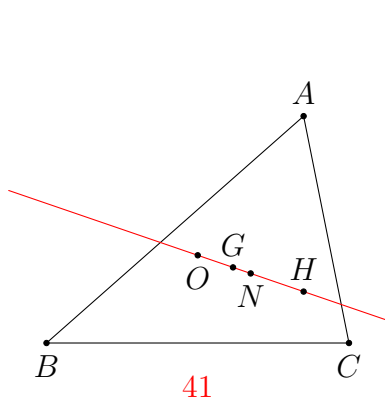
Then the radius of $\odot N$ is $\frac{R}{2}$.

41. (5)(4) Let $\triangle ABC$ be a given triangle.

Then O, G, N, H are collinear in that order and $OG : GN : NH = 2 : 1 : 3$.

(5)(4)(Def) In particular, the line on which O, G, N, H lie is called the *Euler line* (歐拉線) of $\triangle ABC$.

(5)(1) Limiting case: when $\triangle ABC$ is equilateral, then O, G, N, H coincide.

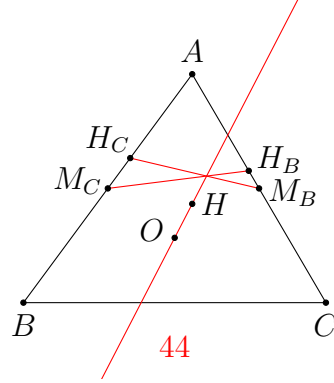
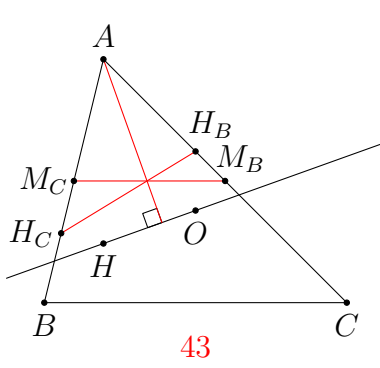


42. (4)(4) Let $\triangle ABC$ be a given triangle.

Then $AM_{AH}M_AO$ and $M_{AH}HM_AO$ are parallelograms ($\sim$).

(5)(4) So $M_{M_{AH}M_A} = N$ and $M_{AH}M_A$ is a diameter of $\odot N$ ($\sim$).

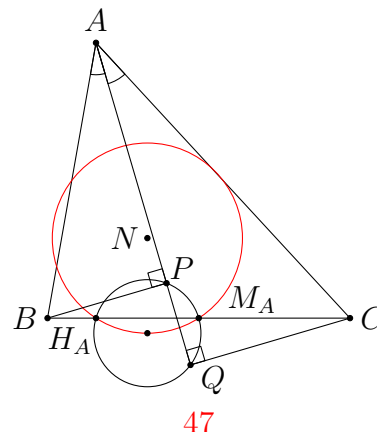
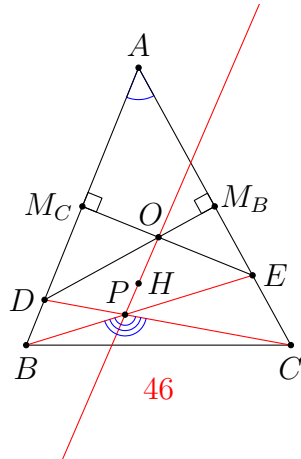
43. (1)(1) Let $\triangle ABC$ be a given triangle.
Then $M_B M_C \cap H_B H_C \cap \perp (A, OH) (\sim)$.



44. (1)(1) Let $\triangle ABC$ be a given triangle.
Then $M_B H_C \cap M_C H_B \cap OH (\sim)$.

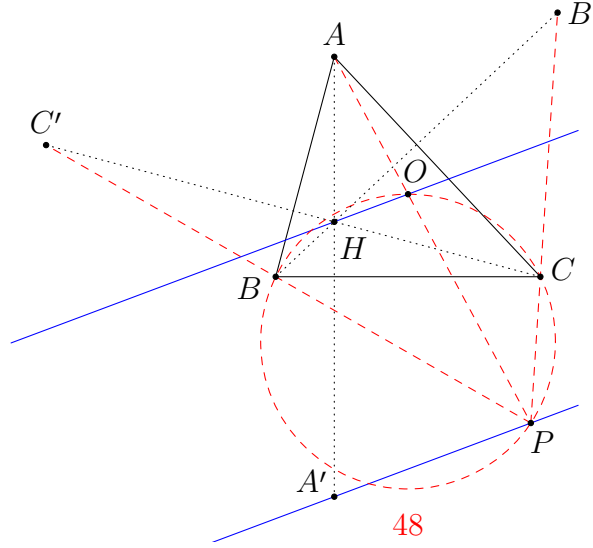
45. (1)(1) Let $\triangle ABC$ be a given triangle and let P be any point.
Then $PA^2 + PB^2 + PC^2 + PH^2 = 4PN^2 + 3R^2$.
(1)(1) So $PA^2 + PB^2 + PC^2 + PH^2$ has the smallest value when $P = N$.

46. (1)(1) Let $\triangle ABC$ be a given triangle. Let $D = M_B O \cap AB$, $E = M_C O \cap AC$.
Then $BE \cap CD \cap OH (\sim)$.
(1)(1) In particular, let the intersection point be P .
Then $\angle BPC = 3A$ (if $A \leq 60^\circ$) or $360^\circ - 3A$ (if $A > 60^\circ$) ($\sim$).

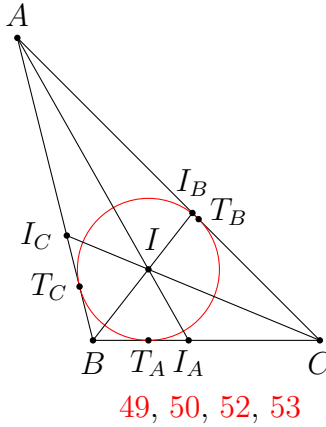


47. (1)(1) (CSMO 2008 Problem 3) Let $\triangle ABC$ be a given triangle. Let $P = \perp B \cap AI$, $Q = \perp C \cap AI$.
Then M_A, P, H_A, Q are concyclic, lying on a circle with centre on $\odot N (\sim)$.

48. (1)(1) Let $\triangle ABC$ be a given triangle. Let $B' = \text{Rf}(B, AC)$, $C' = \text{Rf}(C, AB)$.
Then $C'B \cap B'C \cap AO \cap (BOC) (\sim)$.
(1)(1) In particular, let P be the intersection point and let $A' = \text{Rf}(A, BC)$.
Then $PA' \parallel OH (\sim)$.
(5)(2) Limiting case: when $\angle BAC = 90^\circ$, then $C'B \parallel B'C \parallel AO (\sim)$.



49. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.
Then $AI_A \cap BI_B \cap CI_C = I$, called the *incentre*(内心) of $\triangle ABC$.
50. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.
Then $\odot(I, r)$ touches the sides BC, CA, AB at T_A, T_B, T_C respectively, and is called the *incircle*(内切圆) of $\triangle ABC$.
(5)(5) So $IT_A = IT_B = IT_C = r$ and $I = O_{\triangle T_A T_B T_C}$.



51. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $[IBC] : [ICA] : [IAB] = a : b : c$ and $r = \frac{[ABC]}{s}$.

52. (5)(5) Let $\triangle ABC$ be a given triangle.

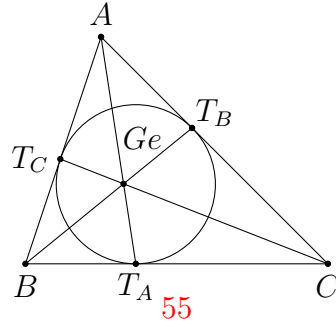
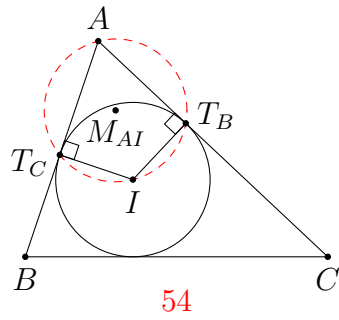
Then $\angle BIC = 90^\circ + \frac{1}{2}A$ ($\sim$).

53. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $AT_B = AT_C = s - a$ ($\sim$).

54. (5)(5) Let $\triangle ABC$ be a given triangle.

Then A, T_C, I, T_B are concyclic, lying on a circle with centre M_{AI} ($\sim$).

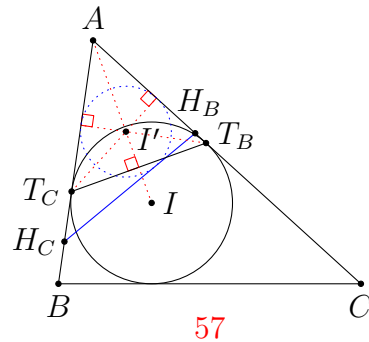
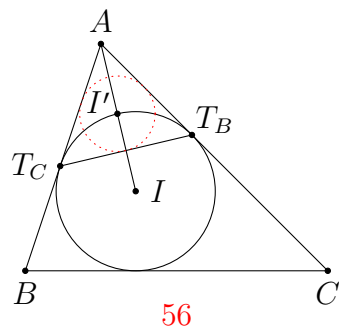


55. (5)(2)(Def) Let $\triangle ABC$ be a given triangle.

Then $AT_A \cap BT_B \cap CT_C = Ge$, called the *Gergonne point* (熱爾崗點) of $\triangle ABC$.

56. (3)(2) Let $\triangle ABC$ be a given triangle. Let $I' = AI \cap \odot I$ (segment AI).

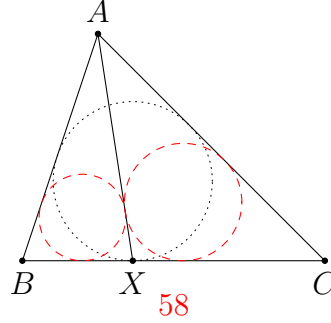
Then $I' = I_{\triangle AT_C T_B}$ ($\sim$) (see also fact 261).



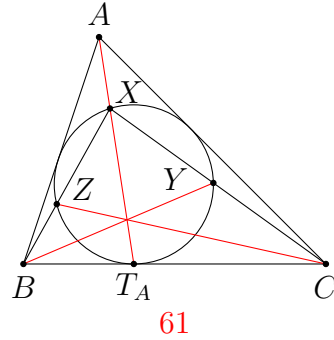
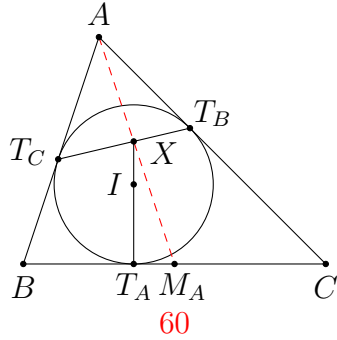
57. (3)(1) Let $\triangle ABC$ be a given triangle. Let $I' = \text{Rf}(I, T_B T_C)$.

Then $I' = H_{\triangle AT_C T_B}$ and $I' = I_{\triangle AH_C H_B}$ ($\sim$).

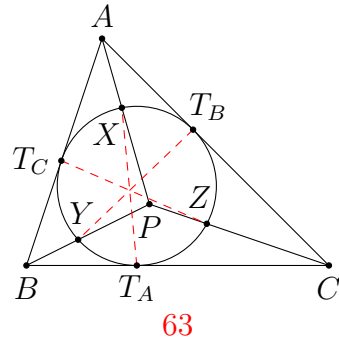
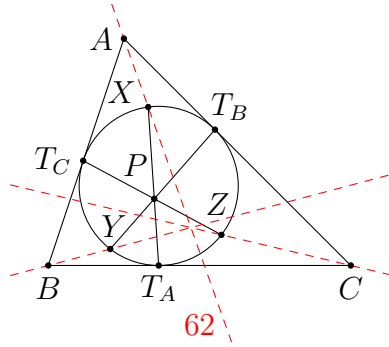
58. (2)(1) In $\triangle ABC$, a point X lies on the side BC .
Then $\odot I_{\triangle ABX}$ is tangent to $\odot I_{\triangle ACX}$ iff $X = T_A$ ($\sim$).



59. (1)(1) Let $\triangle ABC$ be a given triangle and let P be any point.
Then $aPA^2 + bPB^2 + cPC^2 = (a + b + c)PI^2 + abc$.
(1)(1) So $aPA^2 + bPB^2 + cPC^2$ has the smallest value when $P = I$.
60. (4)(3) (NMO 2010 Problem 6) In $\triangle ABC$, let $X = T_A I \cap T_B T_C$.
Then A, X, M_A are collinear ($\sim$).

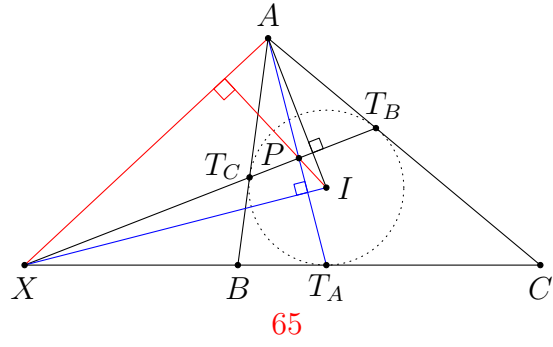
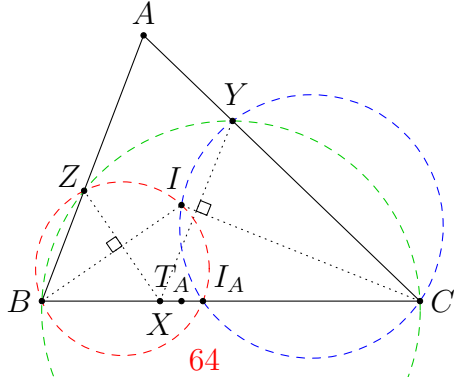


61. (2)(1) In $\triangle ABC$, let $X = AT_A \cap \odot I$ and let $Z, Y = BX, CX \cap \odot I$.
Then $AT_A \cap BY \cap CZ$ ($\sim$).
62. (1)(1) Let $\triangle ABC$ be a given triangle and let $P \neq T_A, T_B, T_C$ be any point. Let $X, Y, Z = PT_A, PT_B, PT_C \cap \odot I$.
Then $AX \cap BY \cap CZ$.
63. (1)(1) (NMO 2011 Problem 2) In $\triangle ABC$, P is an interior point. Let X, Y, Z be the first intersection points of the rays AP, BP, CP and $\odot I$ respectively.
Then $T_A X \cap T_B Y \cap T_C Z$.



64. (1)(1) In $\triangle ABC$, let $X = \text{Rf}(I_A, T_A)$, $Y = \text{Rf}(X, CI)$, $Z = \text{Rf}(X, BI)$ (hence Y, Z lie on the sides AC, AB respectively).

Then C, Y, I, I_A are concyclic, B, Z, I, I_A are concyclic and B, C, Y, Z are concyclic ($\sim$).



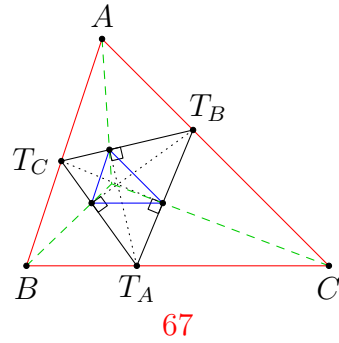
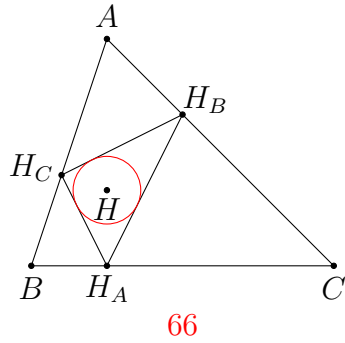
65. (1)(1) Let $\triangle ABC$ be a given triangle. Let $X, P = BC, AT_A \cap T_B T_C$.

Then $P = H_{\triangle AXI}$ ($\sim$).

(1)(1) So $PI \perp AX$ and $AT_A \perp XI$ ($\sim$).

66. (5)(3) Let $\triangle ABC$ be a given triangle.

Then $H = I_{\triangle H_A H_B H_C}$.

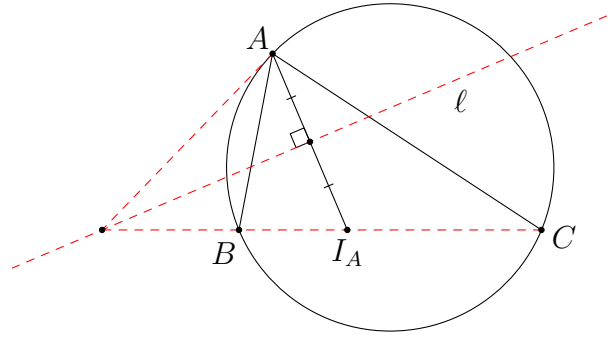


67. (2)(1) Let $\triangle ABC$ be a given triangle.

Then $\triangle ABC$ and the orthic triangle of $\triangle T_A T_B T_C$ are homothetic to each other.

68. (5)(4) Let $\triangle ABC$ be a given triangle. Let ℓ be the perpendicular bisector of AI_A .

Then $BC \cap \ell \cap \tan(A, \odot O) (\sim)$.



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69. (5)(3) (Euler's formula(欧拉公式)) Let $\triangle ABC$ be a given triangle.

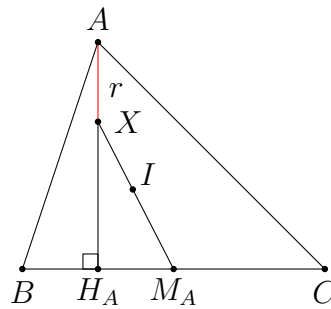
Then $OI^2 = R^2 - 2Rr$.

(5)(2) (Euler's inequality(欧拉不等式)) So $R \geq 2r$.

70. (2)(1) (All-Soviet Union MO 1970 Problem 138) Let $\triangle ABC$ be a given triangle.

Let $X = AH_A \cap M_A I$.

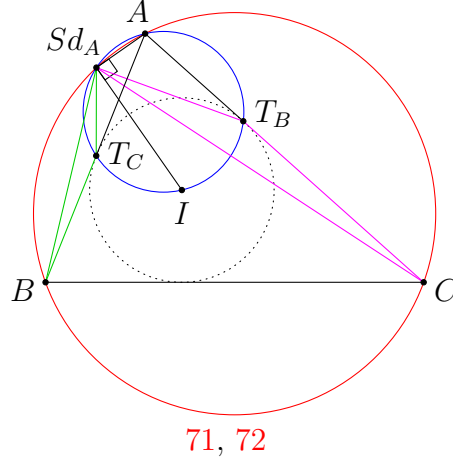
Then $AX = r (\sim)$.



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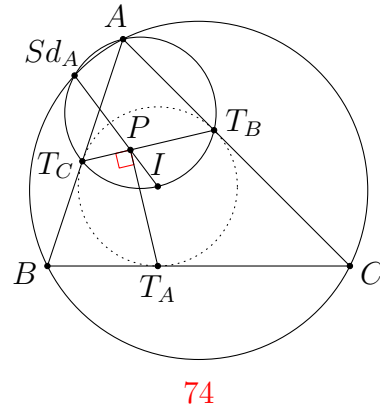
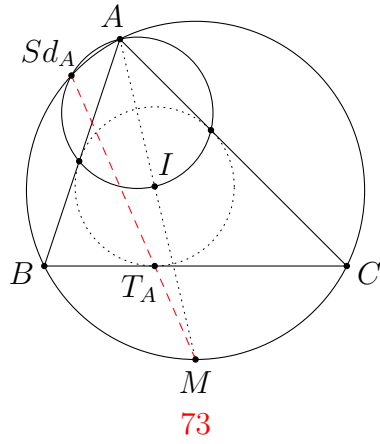
71. (4)(3)(Def) Let $\triangle ABC$ be a given triangle. The *A-Sharky-Devil point* Sd_A of $\triangle ABC$ is defined as the intersection point of (ABC) and $(AT_C IT_B)$ different from $A (\sim)$.

(5)(3) So $\angle ASd_A I = 90^\circ (\sim)$.



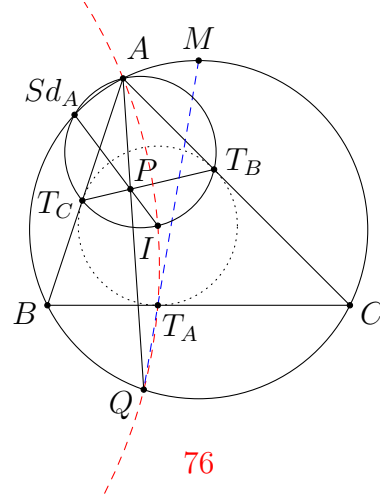
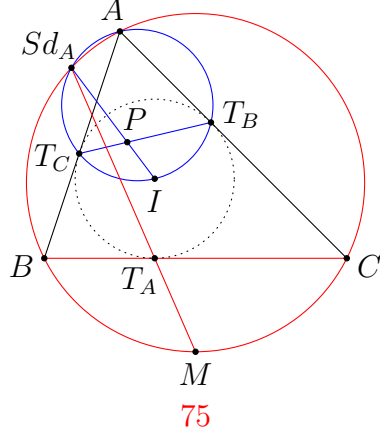
72. (5)(3) Let $\triangle ABC$ be a given triangle.
Then $\triangle Sd_A BT_C \sim \triangle Sd_A CT_B$ ($\sim$).

73. (3)(2) In $\triangle ABC$, let M be the midpoint of $\widehat{BC}$ of $\odot O$ not containing A .
Then M, T_A, Sd_A are collinear ($\sim$).



74. (3)(2) In $\triangle ABC$, let $P = T_B T_C \cap ISd_A$.
Then $T_A P \perp T_B T_C$ ($\sim$).

75. (3)(3) In $\triangle ABC$, let M be the midpoint of $\widehat{BC}$ of $\odot O$ not containing A , and let $P = T_B T_C \cap ISd_A$.
Then $Sd_A BT_A CM \sim Sd_A T_C PT_B I$ ($\sim$).



76. (1)(1) In $\triangle ABC$, let $P = T_B T_C \cap I Sd_A$ and let $Q = AP \cap \odot O$.

Then A, I, T_A, Q are concyclic ($\sim$).

(1)(1) In particular, let M be the midpoint of $\widehat{BAC}$ of $\odot O$.

Then M, T_A, Q are collinear ($\sim$).

77. (3)(1) Let $\triangle ABC$ be a given triangle.

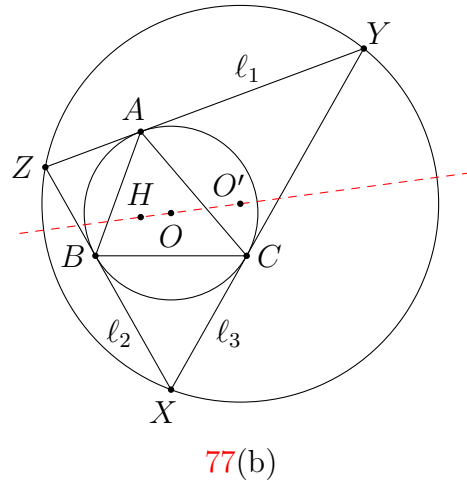
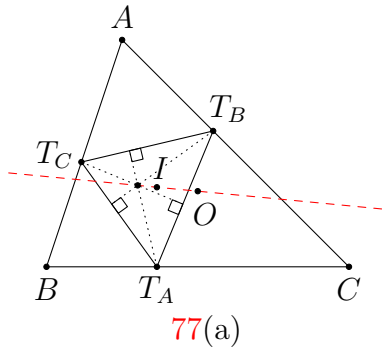
Then $O, I, H_{\triangle T_A T_B T_C}$ are collinear (lying on the Euler line of $\triangle T_A T_B T_C$).

(5)(1) Limiting case: when $\triangle ABC$ is equilateral, then $O, I, H_{\triangle T_A T_B T_C}$ coincide.

(3)(1) So let $\ell_1 = \tan(A, \odot O)$, $\ell_2 = \tan(B, \odot O)$, $\ell_3 = \tan(C, \odot O)$ and let $X = \ell_2 \cap \ell_3$, $Y = \ell_3 \cap \ell_1$, $Z = \ell_1 \cap \ell_2$, $O' = O_{\triangle XYZ}$.

Then O', O, H are collinear (lying on the Euler line of $\triangle ABC$).

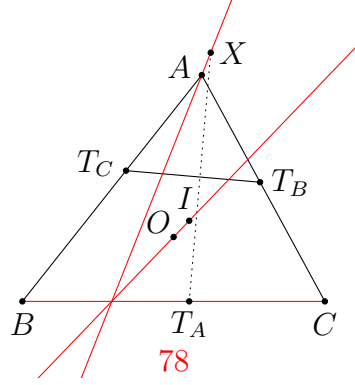
(5)(1) Limiting case: when $\triangle ABC$ is equilateral, then O', O, H coincide.



78. (1)(1) Let $\triangle ABC$ be a given triangle. Let $X = \text{Rf}(T_A, T_B T_C)$.

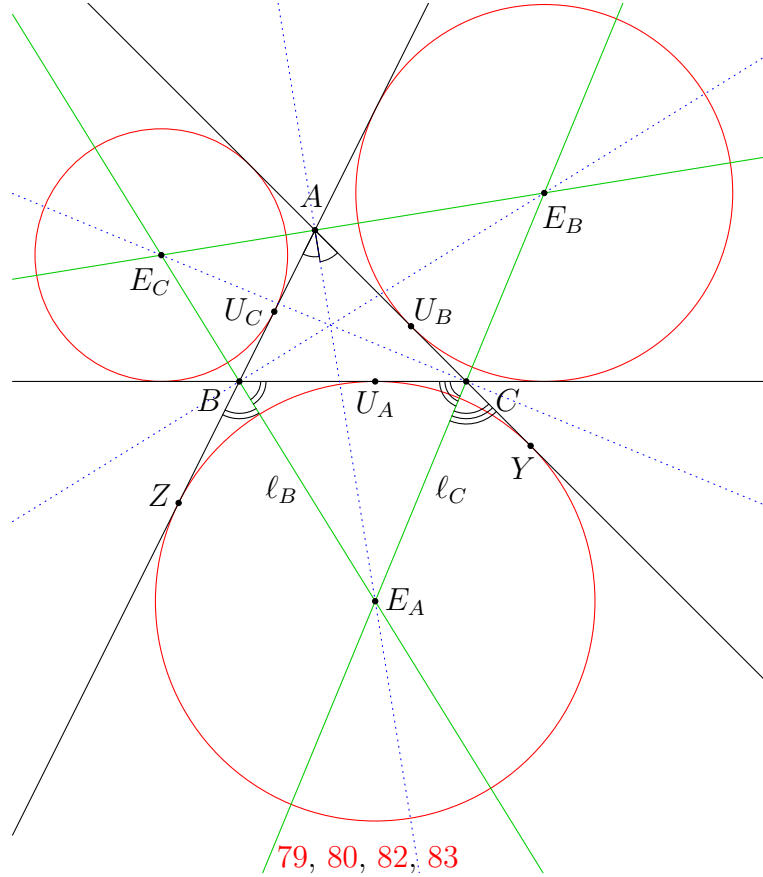
Then $BC \cap OI \cap AX \sim$.

(5)(1) Limiting case: when $\triangle ABC$ is equilateral, then $X = A$ and $I = O$.



79. (5)(5)(Def) Let $\triangle ABC$ be a given triangle. Let ℓ_B, ℓ_C be the external angle bisectors of $\angle B, \angle C$ respectively.

Then $AI_A \cap \ell_B \cap \ell_C = E_A$, called the A -*excentre* (旁心) of $\triangle ABC \sim$.



80. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then $\odot(E_A, r_A)$ touches the side BC at U_A and touches the extension of the sides AC, AB , and is called the A -*excircle* (旁切圓) of $\triangle ABC$ ($\sim$).

(5)(5) So $E_A U_A = r_A$ ($\sim$).

81. (5)(3) Let $\triangle ABC$ be a given triangle.

Then $[E_A CB] : [E_A CA] : [E_A AB] = a : b : c$ and $r_A = \frac{[ABC]}{s - a}$ ($\sim$).

82. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $\angle BE_A C = 90^\circ - \frac{1}{2}A$ and $\angle AE_A B = \frac{1}{2}C$ ($\sim$).

83. (5)(5) Let $\triangle ABC$ be a given triangle. Let Z, Y be the contact points of $\odot E_A$ and the sidelines AB, AC respectively.

Then $BU_A = s - c$, $CU_A = s - b$ and $AZ = AY = s$ ($\sim$).

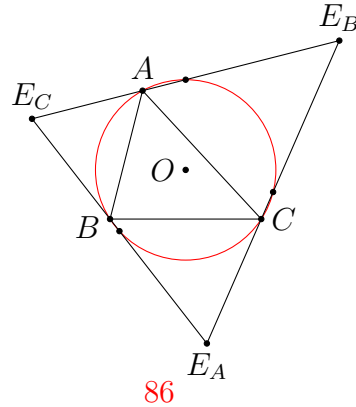
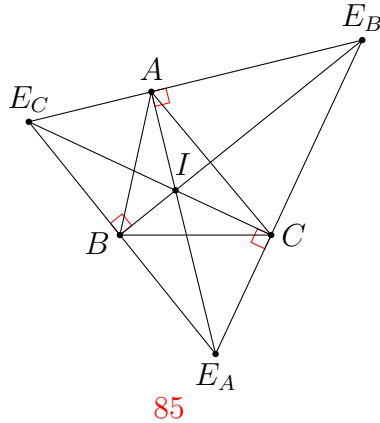
84. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $AB \cdot AC = AI \cdot AE_A$ ($\sim$).

85. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $AI_A \perp E_B E_C$ ($\sim$).

(5)(4) So $I = H_{\triangle E_A E_B E_C}$.



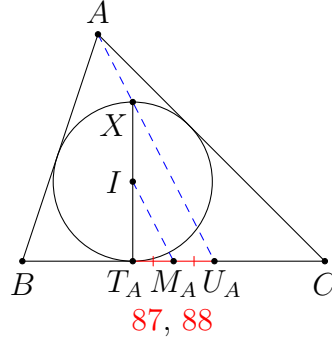
86. (5)(3) Let $\triangle ABC$ be a given triangle.

Then $\odot O = \odot N_{\triangle E_A E_B E_C}$.

(5)(3) So $O = N_{\triangle E_A E_B E_C}$.

87. (5)(5) Let $\triangle ABC$ be a given triangle.

Then $M_{T_A U_A} = M_A (\sim)$.



88. (5)(4) Let $\triangle ABC$ be a given triangle. Let $X = T_A I \cap \odot I$.

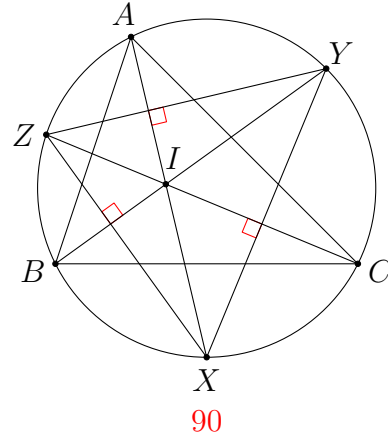
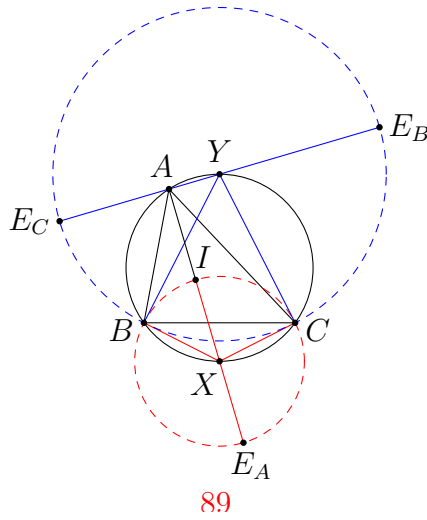
Then A, X, U_A are collinear ($\sim$).

(4)(3) So $IM_A \parallel AU_A$.

89. (5)(5) Let $\triangle ABC$ be a given triangle. Let $X, Y = AI_A, E_B E_C \cap \odot O$.

Then $XB = XC = XI = XE_A$ and $YB = YC = YE_B = YE_C (\sim)$.

(5)(5) So X, Y are the midpoints of $\widehat{BC}$ of $\odot O (\sim)$.



90. (5)(4) In $\triangle ABC$, let X, Y, Z be the midpoints of $\widehat{BC}, \widehat{CA}, \widehat{AB}$ of $\odot O$ not containing the opposite vertices.

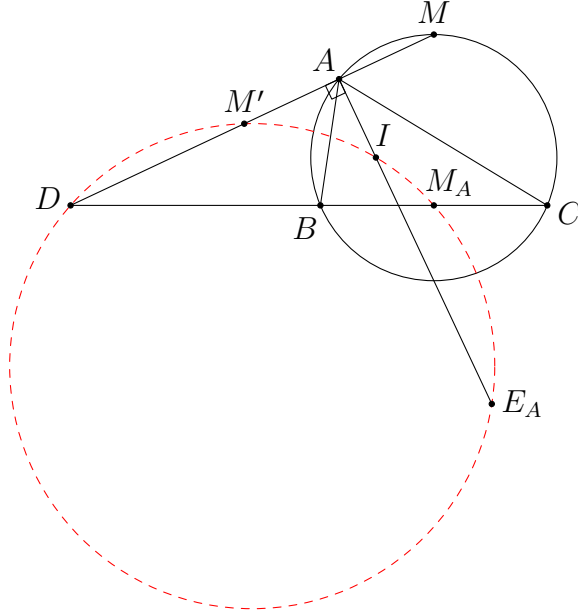
Then $AX \perp YZ (\sim)$.

(5)(4) So $I = H_{\triangle XYZ}$ and YZ is the perpendicular bisector of $AI (\sim)$.

91. (3)(3) In $\triangle ABC$, let M be the midpoint of $\widehat{BAC}$ of $\odot O$. Let $D = AM \cap BC$ and let $M' = \text{Rf}(M, A)$.

Then D, E_A, M_A, I, M' are concyclic ($\sim$).

(3)(3) So $AM \cdot AD = AI \cdot AE_A$ ($\sim$).

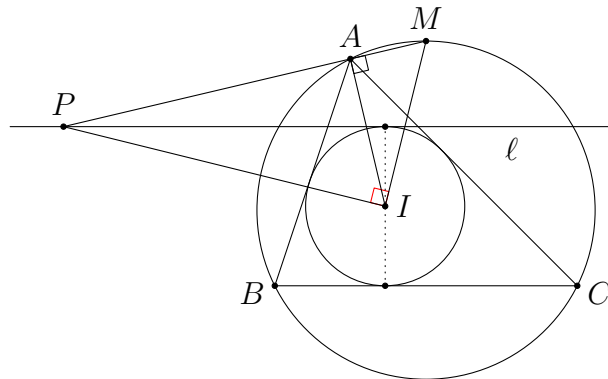


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92. (1)(1) In $\triangle ABC$, let M be the midpoint of $\widehat{BAC}$ of $\odot O$. Let $\ell \neq BC$ be the tangent to $\odot I$ which is parallel to BC , and let $P = AM \cap \ell$.

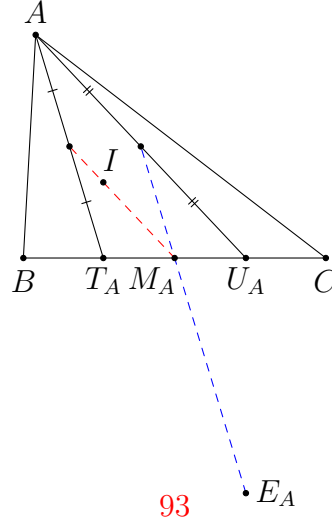
Then $\angle MIP = 90^\circ$ ($\sim$).

(1)(1) So $\triangle AIP \sim \triangle AMI$ and $AM \cdot AP = AI^2$ ($\sim$).

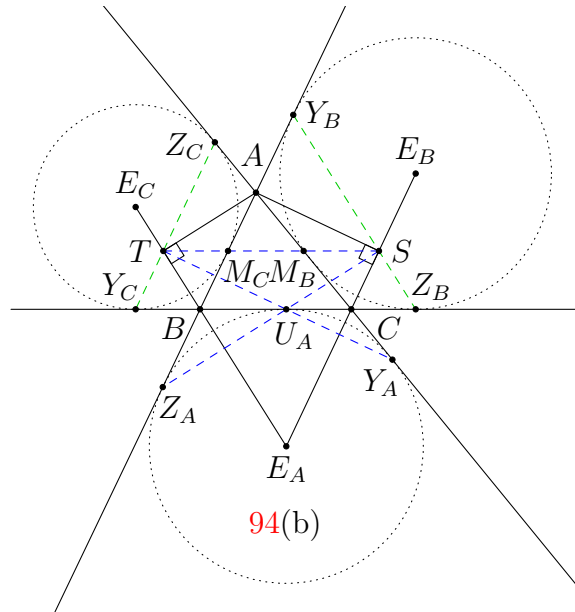
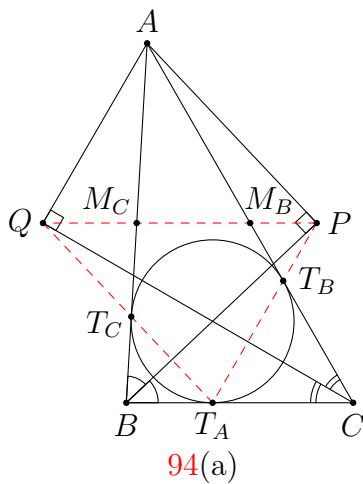


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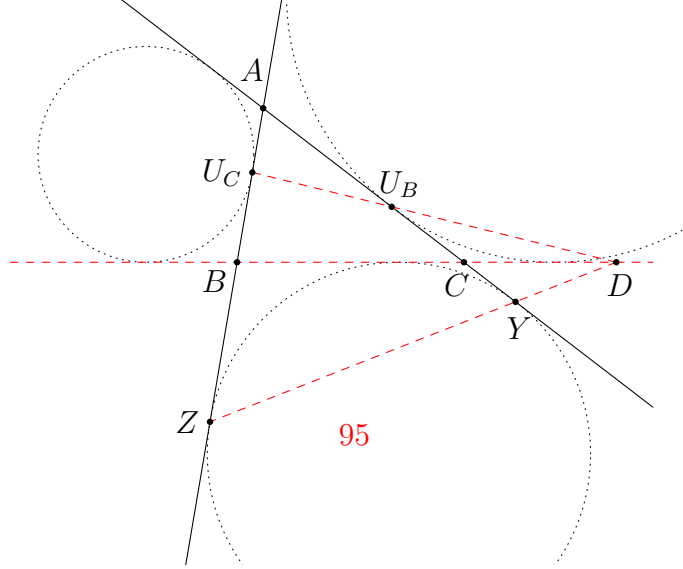
93. (1)(1) (All-Soviet Union MO 1968 Problem 112) Let $\triangle ABC$ be a given triangle. Then I, M_A, M_{AT_A} are collinear ($\sim$).
 (1)(1) Similarly, E_A, M_A, M_{AU_A} are collinear ($\sim$).



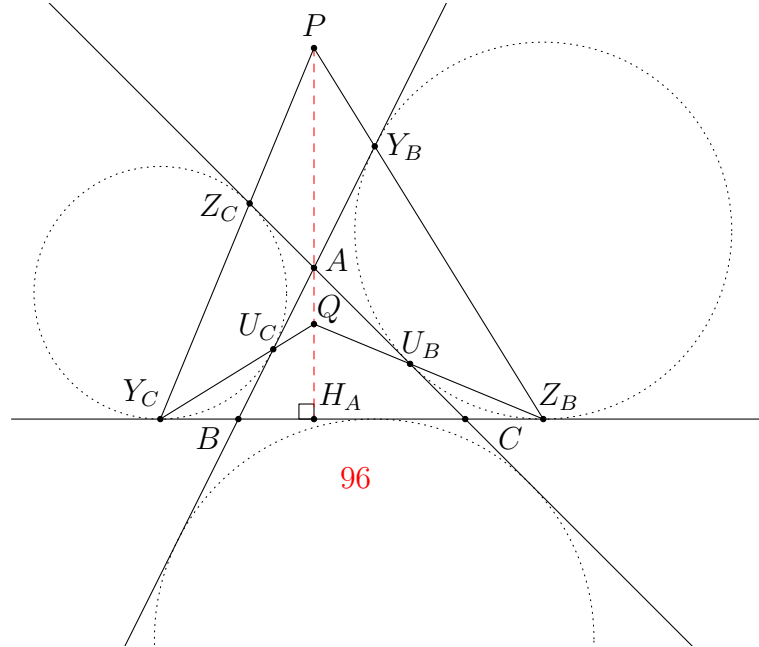
94. (Iran lemma(伊朗引理)) (4)(4) In $\triangle ABC$, let $P, Q = \perp A \cap BI_B, CI_C$. Then T_A, T_B, P are collinear, T_A, T_C, Q are collinear, and P, M_B, M_C, Q are collinear ($\sim$).
 (3)(3) Similarly, let $S = \perp A \cap E_A E_B$, $T = \perp A \cap E_A E_C$, $Z_A, Y_A = \odot E_A \cap AB, AC$, $Z_B, Y_B = \odot E_B \cap BC, BA$, $Z_C, Y_C = \odot E_C \cap CA, CB$.
 Then Z_A, U_A, S are collinear, Y_A, U_A, T are collinear, Z_B, S, Y_B are collinear, Z_C, T, Y_C are collinear, and S, M_B, M_C, T are collinear ($\sim$).



95. (1)(1) Let $\triangle ABC$ be a given triangle. Let $Z, Y = \odot E_A \cap AB, AC$.
Then $BC \cap U_C U_B \cap ZY (\sim)$.



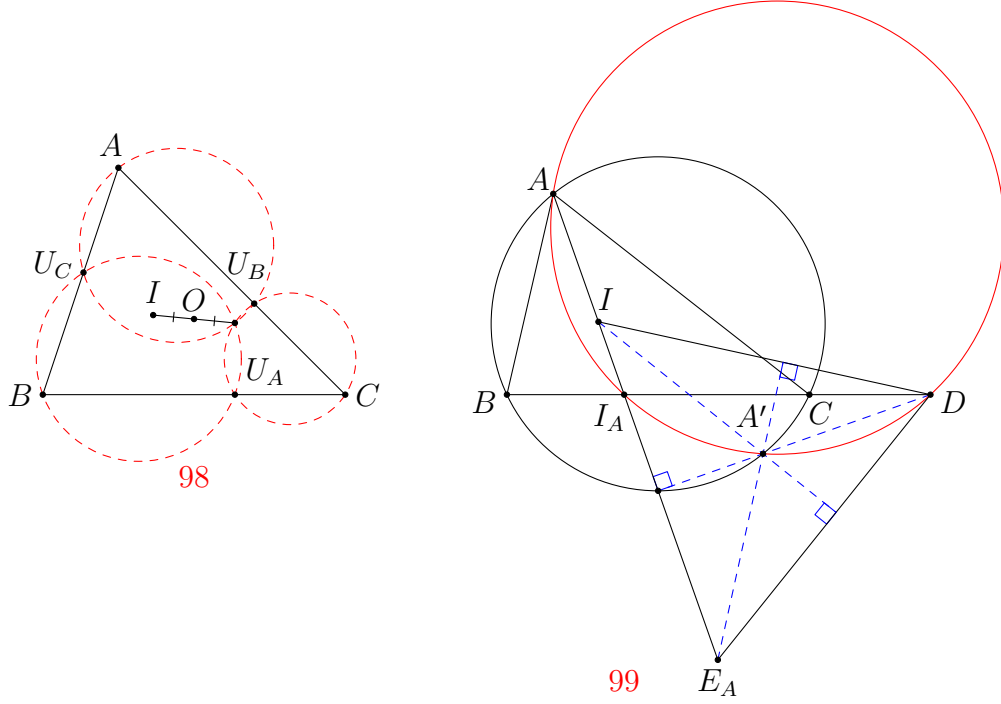
96. (1)(1) In $\triangle ABC$, let $Z_B, Y_B = \odot E_B \cap BC, BA$, $Z_C, Y_C = \odot E_C \cap CA, CB$,
 $P = Z_B Y_B \cap Z_C Y_C$, $Q = Z_B U_B \cap Y_C U_C$.
Then P, A, Q, H_A are collinear ($\sim$).



97. (2)(2) Let $\triangle ABC$ be a given triangle.
Then $M_A I_A \cdot M_A H_A = M_A T_A^2 = M_A U_A^2 (\sim)$.

98. (2)(1) Let $\triangle ABC$ be a given triangle.

Then $(AU_CU_B) \cap (BU_AU_C) \cap (CU_BU_A) = \text{Rf}(I, O)$.

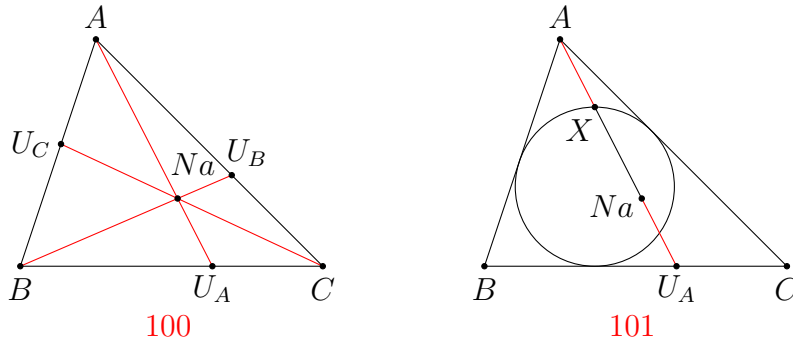


99. (1)(1) In $\triangle ABC$, let $A' = \text{Rf}(A, O)$ and $D = BC \cap (AI_A A')$.

Then $D = H_{\triangle IE_A A'}$.

100. (5)(1)(Def) Let $\triangle ABC$ be a given triangle.

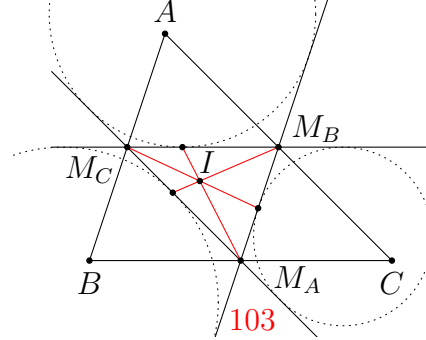
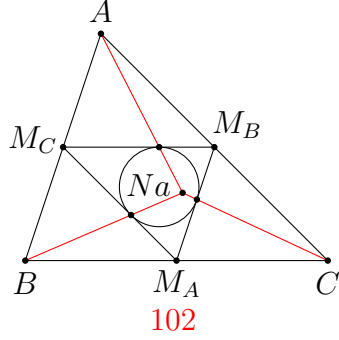
Then $AU_A \cap BU_B \cap CU_C = Na$, called the *Nagel point* (奈格爾點) of $\triangle ABC$.



101. (2)(1) Let $\triangle ABC$ be a given triangle. Let X be the first intersection point of the ray AU_A and $\odot I$.

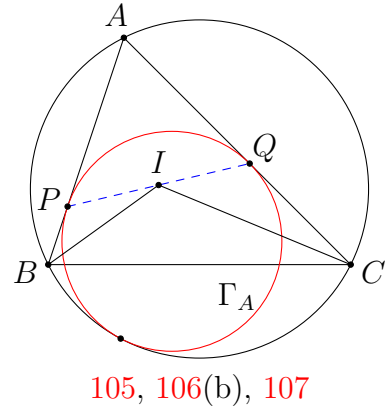
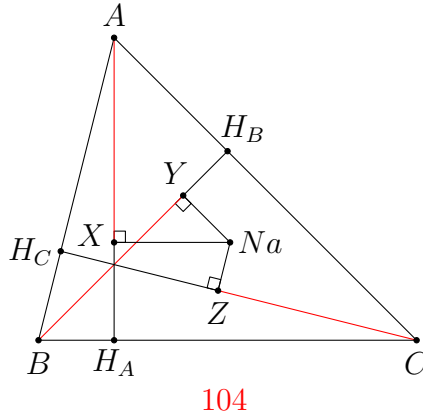
Then $AX = NaU_A$ ($\sim$).

102. (1)(1) Let $\triangle ABC$ be a given triangle. Let $\triangle = \triangle M_A M_B M_C$.
Then $A(T_{M_A})_{\triangle} \cap B(T_{M_B})_{\triangle} \cap C(T_{M_C})_{\triangle} = Na$.

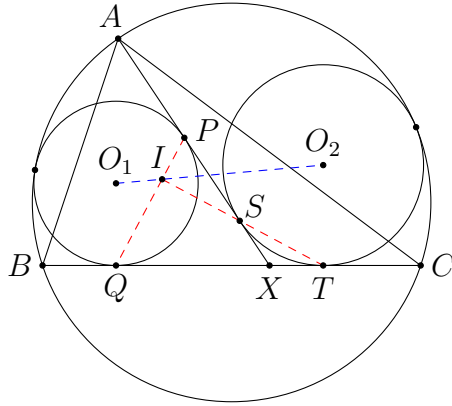


103. (2)(1) Let $\triangle ABC$ be a given triangle.
Then $I = Na_{\triangle M_A M_B M_C}$.

104. (1)(1) In $\triangle ABC$, let $X, Y, Z \perp Na \cap AH_A, BH_B, CH_C$.
Then $AX = BY = CZ$.



105. (4)(3)(Def) Let $\triangle ABC$ be a given triangle.
Then there exists a unique circle Γ_A tangent to the sides AB, AC and internally tangent to $\odot O$, called the A -*mixtilinear incircle* (偽内切圓) of $\triangle ABC$ ($\sim$).
106. (4)(3) (Y. Sawayama's lemma (澤山定理)) In $\triangle ABC$, a point X lies on the side BC . Let $\odot O_1$ be a circle internally tangent to $\odot O$ and tangent to the segments AX, BX at P, Q respectively. Let $\odot O_2$ be a circle internally tangent to $\odot O$ and tangent to the segments AX, CX at S, T respectively.
Then $PQ \cap ST = I$ ($\sim$).
- (4)(3) (Thébaud's theorem (泰博定理)) In addition, O_1, I, O_2 are collinear.
- (4)(3) Limiting case: when $X = C$ (which means $\odot O_1$ is the C -mixtilinear incircle), P, I, Q are collinear.



106(a)

107. (3)(2) In $\triangle ABC$, let $P, Q = AB, AC \cap \Gamma_A$.

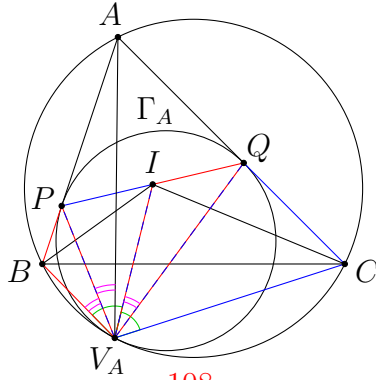
Then $\triangle BPI \sim \triangle BIC \sim \triangle IQC$ ($\sim$).

108. (3)(2) In $\triangle ABC$, let $P, Q = AB, AC \cap \Gamma_A$.

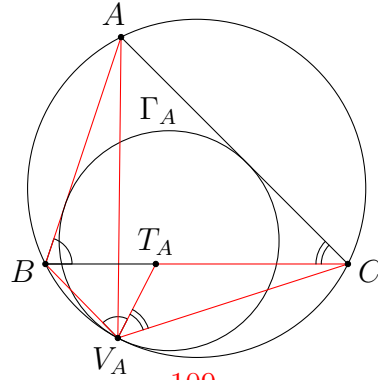
Then $\triangle V_A BP \sim \triangle V_A IQ$, $\triangle V_A CQ \sim \triangle V_A IP$ and $\triangle V_A BI \sim \triangle V_A PQ \sim \triangle V_A IC$ ($\sim$).

(3)(1) So $\angle BV_A I = \angle CV_A I$ ($\sim$).

(3)(1) In particular, $\angle AV_A P = \angle BV_A P = \angle IV_A Q$ ($\sim$).



108



109

109. (3)(2) Let $\triangle ABC$ be a given triangle.

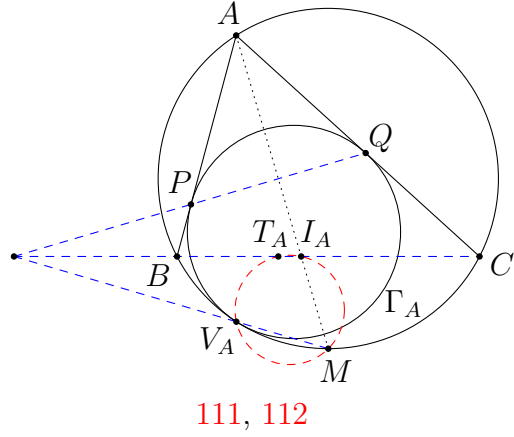
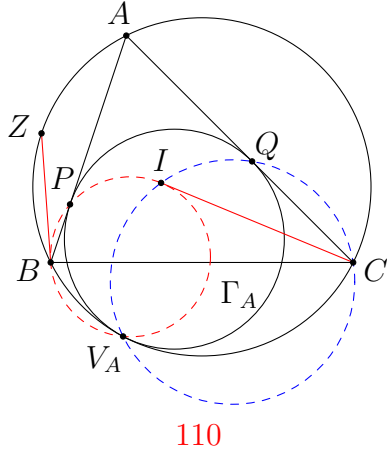
Then $\triangle ABV_A \sim \triangle CT_A V_A$ and $\triangle ACV_A \sim \triangle BT_A V_A$ ($\sim$).

(3)(1) So $\angle BV_A T_A = B$ and $\angle CV_A T_A = C$ ($\sim$).

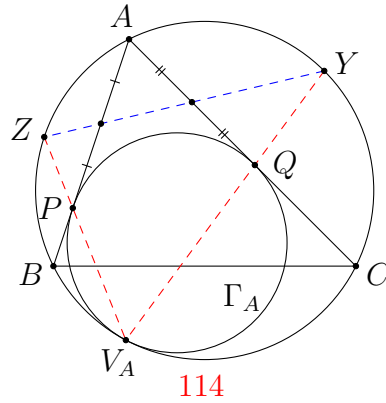
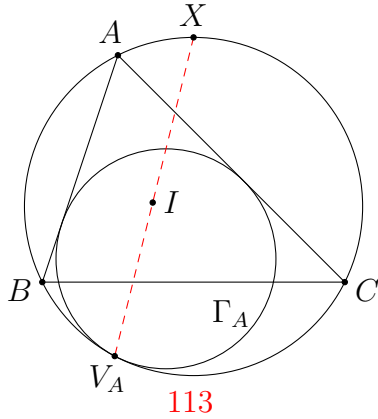
110. (3)(2) In $\triangle ABC$, let $P, Q = AB, AC \cap \Gamma_A$. Let Z be the midpoint of $\widehat{AB}$ of $\odot O$ not containing C .

Then B, P, I, V_A are concyclic, C, Q, I, V_A are concyclic ($\sim$).

(2)(1) In particular, CI and BZ are tangent to $(BPIV_A)$ ($\sim$).



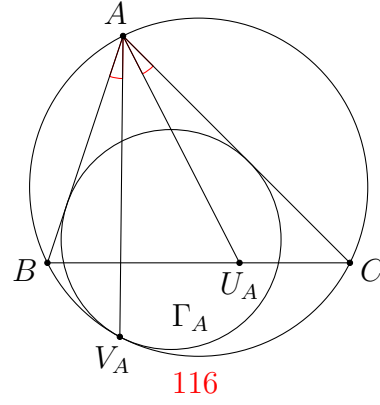
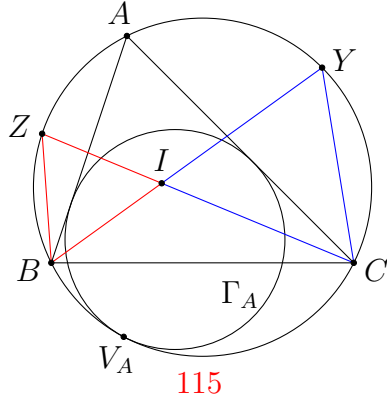
111. (2)(1) In $\triangle ABC$, let M be the midpoint of $\widehat{BC}$ of $\odot O$ not containing A .
Then V_A, T_A, I_A, M are concyclic ($\sim$).
112. (2)(1) In $\triangle ABC$, let $P, Q = AB, AC \cap \Gamma_A$. Let M be the midpoint of $\widehat{BC}$ of $\odot O$ not containing A .
Then $PQ \cap BC \cap V_A M$ ($\sim$).
113. (3)(2) In $\triangle ABC$, let X be the midpoint of $\widehat{BAC}$ of $\odot O$.
Then V_A, I, X are collinear ($\sim$).



114. (3)(3) In $\triangle ABC$, let $P, Q = AB, AC \cap \Gamma_A$. Let Y, Z be the midpoints of $\widehat{AC}, \widehat{AB}$ of $\odot O$ not containing the opposite vertices.
Then V_A, P, Z are collinear, V_A, Q, Y are collinear, Z, M_{AP}, M_{AQ}, Y are collinear ($\sim$).

115. (2)(1) In $\triangle ABC$, let Y, Z be the midpoints of $\widehat{AC}, \widehat{AB}$ of $\odot O$ not containing the opposite vertices.

Then $V_A B Z I \sim V_A I Y C$ ($\sim$).

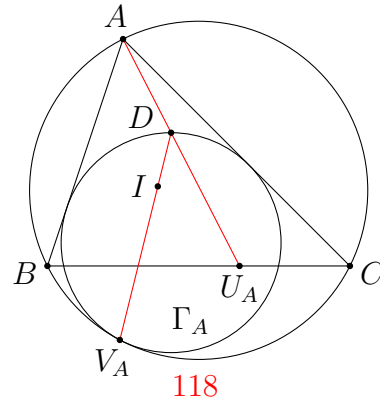
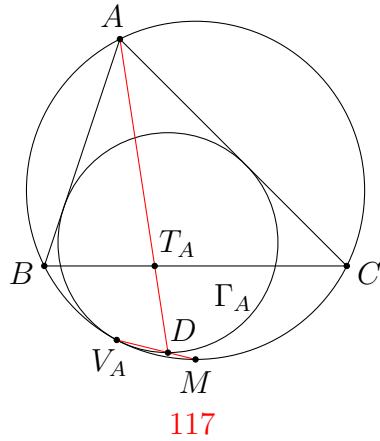


116. (4)(3) Let $\triangle ABC$ be a given triangle.

Then $\angle BAV_A = \angle CAU_A$ ($\sim$).

117. (1)(1) In $\triangle ABC$, let M be the midpoint of $\widehat{BC}$ of $\odot O$ not containing A . Let $D = AT_A \cap MV_A$.

Then D lies on Γ_A ($\sim$).

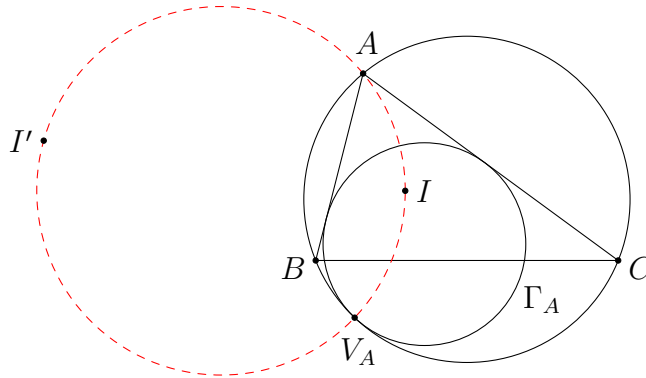


118. (1)(1) In $\triangle ABC$, let $D = AU_A \cap IV_A$.

Then D lies on Γ_A ($\sim$).

119. (1)(1) In $\triangle ABC$, let $I' = \text{Inv}(I, \odot O)$.

Then A, I, V_A, I' are concyclic ($\sim$).



119

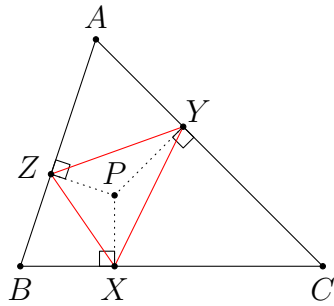
120. (5)(2)(Def) Let $\triangle ABC$ be a given triangle and let $P \neq A, B, C$ be any point. Let $X, Y, Z = \perp P \cap BC, CA, AB$.

Then $\triangle XYZ$ is called the *pedal triangle*(垂足三角形) of P w.r.t. $\triangle ABC$.

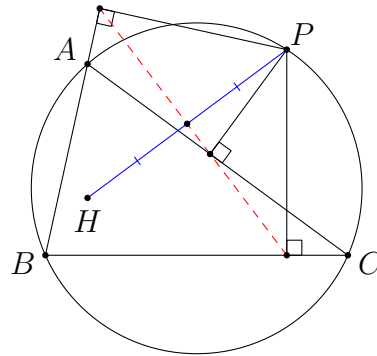
(5)(5) In particular, the pedal triangle of H w.r.t. $\triangle ABC$ is $\triangle H_A H_B H_C$ (and so it has the same Chinese name as the general case).

(5)(5) The pedal triangle of O w.r.t. $\triangle ABC$ is $\triangle M_A M_B M_C$.

(5)(5) The pedal triangle of I w.r.t. $\triangle ABC$ is $\triangle T_A T_B T_C$.



120



121, 122

121. (5)(3) Let $\triangle ABC$ be a given triangle and let $P \neq A, B, C$ be any point.

Then the pedal triangle of P w.r.t. $\triangle ABC$ is a straight line iff P lies on $\odot O$.

(5)(3) In particular, when P lies on $\odot O$, the straight line is called the *Simson line*(西姆松线) of P w.r.t. $\triangle ABC$.

122. (5)(2) In $\triangle ABC$, let P be a point on $\odot O$.

Then the Simson line of P w.r.t. $\triangle ABC$ passes through M_{PH} .

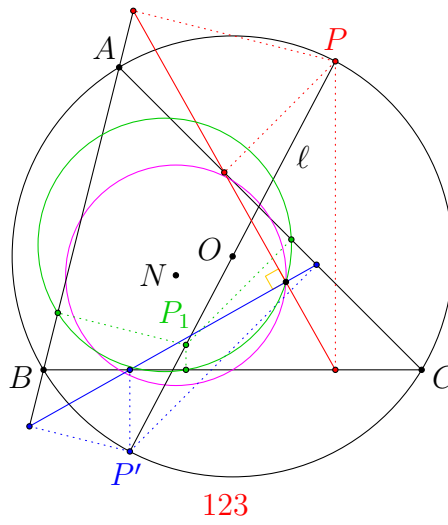
123. (2)(1) (Griffiths' theorem) Let $\triangle ABC$ be a given triangle and let $\ell(O)$ be a fixed line. A movable point $P \neq A, B, C$ lies on $\ell(O)$. Let $\triangle XYZ$ be the pedal triangle of P w.r.t. $\triangle ABC$.

Then (XYZ) (or the line XYZ for the degenerate case) passes through a fixed point on $\odot N$.

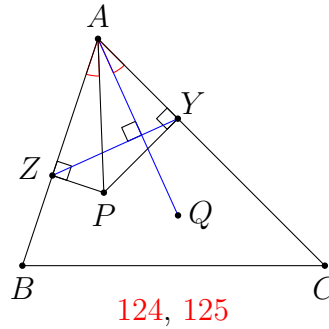
(1)(1) So let PP' be a diameter of $\odot O$.

Then the Simson lines of P and P' w.r.t. $\triangle ABC$ intersect on $\odot N$.

(1)(1) In particular, the Simson lines of P and P' w.r.t. $\triangle ABC$ are perpendicular.



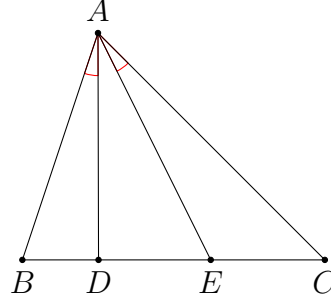
124. (4)(5)(Def) Let $\triangle ABC$ be a given triangle. Let $P, Q \neq A$ be any points both lying inside $\angle BAC$ or both lying outside $\angle BAC$ such that $\angle BAP = \angle CAQ$. Then AP and AQ are called *isogonal lines* (等角線) (or simply *isogonals*) of each other w.r.t. $\angle BAC$.



125. (4)(4) Let $\triangle ABC$ be a given triangle and let AP, AQ be isogonals w.r.t. $\angle BAC$. Let $Z, Y = \perp P \cap AB, AC$. Then $YZ \perp AQ$ ($\sim$).

126. (4)(3) Let $\triangle ABC$ be a given triangle and let D, E be points on the side BC .

Then AD, AE are isogonals w.r.t. $\angle BAC$ iff $\frac{DB}{DC} \cdot \frac{EB}{EC} = \frac{AB^2}{AC^2}$.

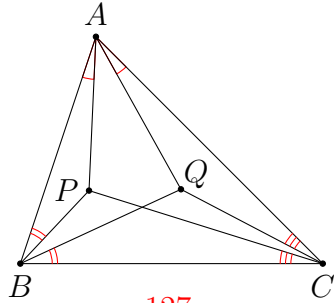


126

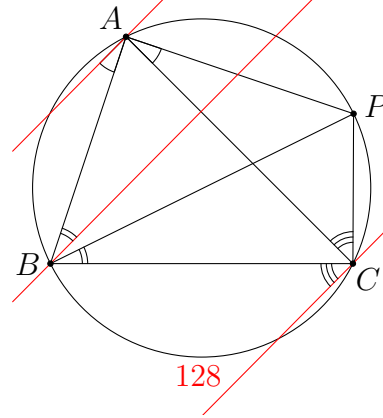
127. (4)(3)(Def) Let $\triangle ABC$ be a given triangle and let P be any point not lying on the lines BC, CA, AB or $\odot O$.

Then the lines $\text{Rf}(AP, AI_A)$, $\text{Rf}(BP, BI_B)$, $\text{Rf}(CP, CI_C)$ are concurrent at a point $Q = \text{Iso}(P)$, called the *isogonal conjugate* (等角共軛) of P w.r.t. $\triangle ABC$.

(5)(5) In particular, $\text{Iso}(I) = I$, $\text{Iso}(H) = O$, $\text{Iso}(O) = H$, $\text{Iso}(E_A) = E_A$, $\text{Iso}(E_B) = E_B$, $\text{Iso}(E_C) = E_C$.



127



128

128. (4)(1) Let $\triangle ABC$ be a given triangle and let P be any point not lying on the lines BC, CA, AB .

Then the lines $\text{Rf}(AP, AI_A)$, $\text{Rf}(BP, BI_B)$, $\text{Rf}(CP, CI_C)$ are parallel iff P lies on $\odot O$.

129. (4)(3) Let $\triangle ABC$ be a given triangle and let P be any point not lying on the lines BC, CA, AB or $\odot O$.

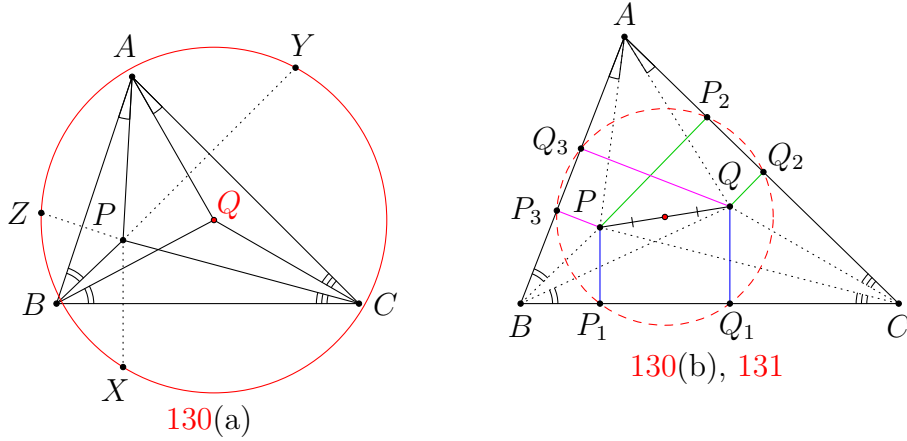
Then $\text{Iso}(\text{Iso}(P)) = P$.

130. (4)(3) Let $\triangle ABC$ be a given triangle and let P, Q be a pair of isogonal conjugates w.r.t. $\triangle ABC$. Let $X = \text{Rf}(P, BC)$, $Y = \text{Rf}(P, CA)$, $Z = \text{Rf}(P, AB)$.

Then $Q = O_{\triangle XYZ}$.

- (4)(2) So let $\triangle P_1P_2P_3$ and $\triangle Q_1Q_2Q_3$ be the pedal triangles of P and Q w.r.t. $\triangle ABC$.

Then $P_1, P_2, P_3, Q_1, Q_2, Q_3$ are concyclic, lying on a circle with centre M_{PQ} .

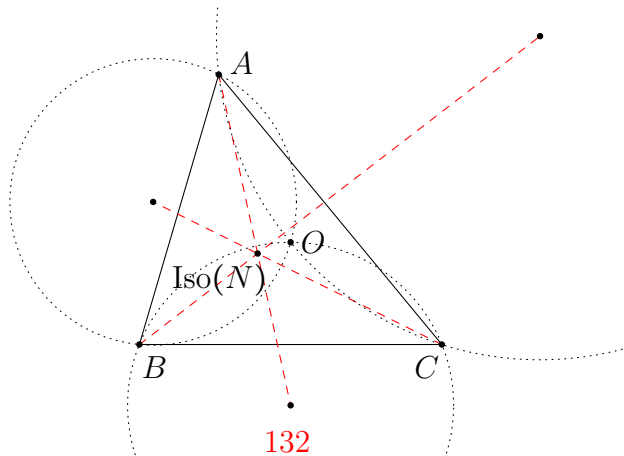


131. (2)(1) Let $\triangle ABC$ be a given triangle and let P, Q be a pair of isogonal conjugates w.r.t. $\triangle ABC$. Let $\triangle P_1P_2P_3$ and $\triangle Q_1Q_2Q_3$ be the pedal triangles of P and Q w.r.t. $\triangle ABC$.

Then $PP_1 \cdot QQ_1 = PP_2 \cdot QQ_2 = PP_3 \cdot QQ_3$.

132. (1)(1) (Kosnita theorem) Let $\triangle ABC$ be a given triangle such that $\text{Iso}(N)$ exists.

Then $AO_{\triangle BOC} \cap BO_{\triangle COA} \cap CO_{\triangle AOB} = \text{Iso}(N)$.



133. (1)(1) Let $\triangle ABC$ be a given triangle.

Then the centre of internal homothety (i.e. with negative ratio) of $\odot I$ and $\odot O$ is $\text{Iso}(Ge)$.

134. (1)(1) Let $\triangle ABC$ be a given triangle.

Then the centre of external homothety (i.e. with positive ratio) of $\odot I$ and $\odot O$ is $\text{Iso}(Na)$.

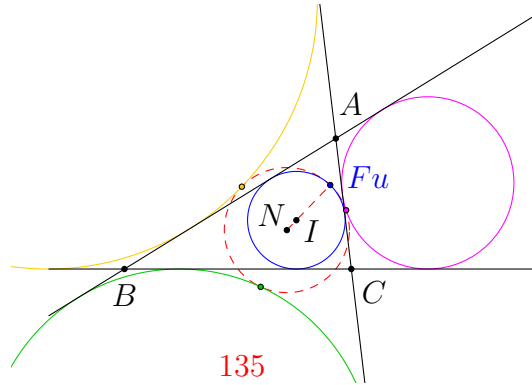
135. (1)(1)(Def) (Fontené theorem) In $\triangle ABC$, let $P \neq A, B, C$ be any point. Let $\triangle XYZ$ be the pedal triangle of P w.r.t. $\triangle ABC$.

Then (XYZ) is tangent to $\odot N$ iff $O, P, \text{Iso}(P)$ are collinear. The locus of all such points P is called the *M'Cay cubic*.

(5)(5) Limiting case: when $P = H, O$, then $(XYZ) = \odot N$.

(5)(2)(Def) (Feuerbach theorem(費爾巴哈定理)) In particular, $\odot N$ is tangent to $\odot I$ (internally), $\odot E_A$, $\odot E_B$ and $\odot E_C$ (externally). The point $Fu = \odot N \cap \odot I$ is called the *Feuerbach point*(費爾巴哈點) of $\triangle ABC$.

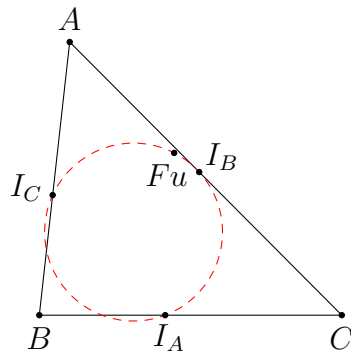
(5)(2) So N, I, Fu are collinear.



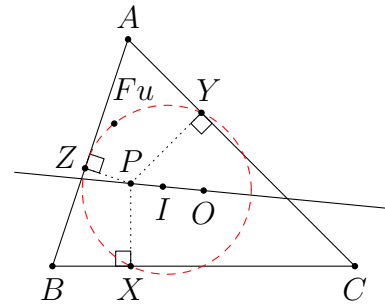
135

136. (1)(1) Let $\triangle ABC$ be a given triangle.

Then I_A, I_B, I_C, Fu are concyclic.



136



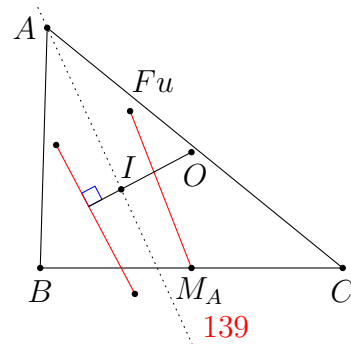
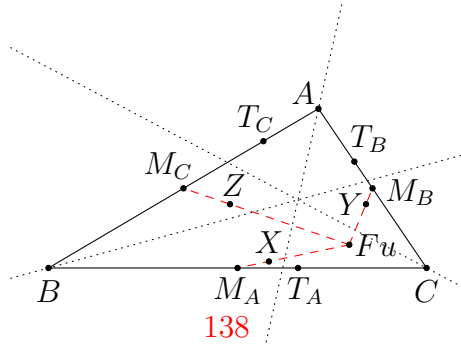
137

137. (1)(1) In $\triangle ABC$, let P be any point on the line OI . Let $\triangle XYZ$ be the pedal triangle of P w.r.t. $\triangle ABC$.

Then X, Y, Z, Fu are concyclic.

138. (1)(1) (IMO 1982 Problem 2) In a scalene $\triangle ABC$, let $X = \text{Rf}(T_A, AI_A)$, $Y = \text{Rf}(T_B, BI_B)$, $Z = \text{Rf}(T_C, CI_C)$.

Then $M_AX \cap M_BY \cap M_CZ = Fu$.

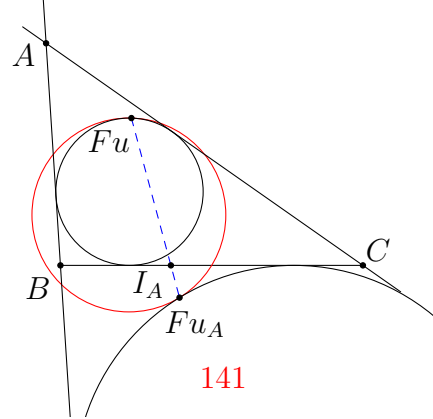
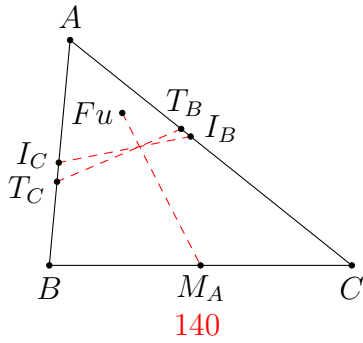


139. (1)(1) Let $\triangle ABC$ be a given triangle.

Then $\text{Rf}(M_A Fu, AI_A) \perp OI$ ($\sim$).

140. (1)(1) Let $\triangle ABC$ be a given triangle.

Then $T_B T_C \cap I_B I_C \cap M_A Fu$ ($\sim$).

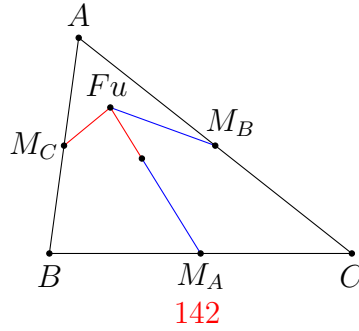


141. (2)(1) Let $\triangle ABC$ be a given triangle. Let $Fu_A = \odot N \cap \odot E_A$.

Then Fu, I_A, Fu_A are collinear ($\sim$).

142. (1)(1) In $\triangle ABC$, let $d_A = FuM_A$, $d_B = FuM_B$, $d_C = FuM_C$.

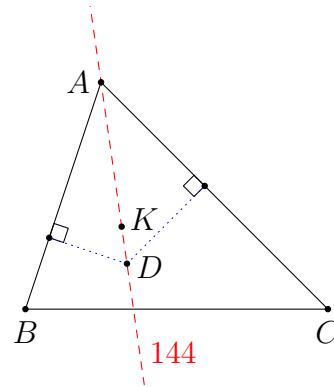
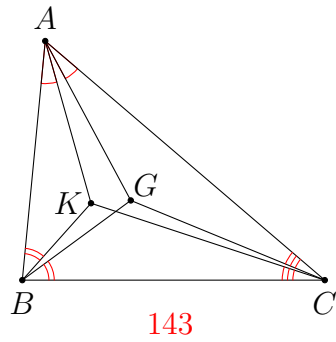
Then $\max\{d_A, d_B, d_C\} = \text{sum of the smallest two of } d_A, d_B, d_C$.



143. (5)(5)(Def) Let $\triangle ABC$ be a given triangle.

Then the isogonal line of AM_A is called the A -*symmedian* (類似中線) of $\triangle ABC$ ($\sim$).

(5)(2)(Def) In particular, the three symmedians are concurrent at $K = \text{Iso}(G)$, called the *symmedian point* (類似重心) or *Lemoine point* (萊莫恩點) of $\triangle ABC$.

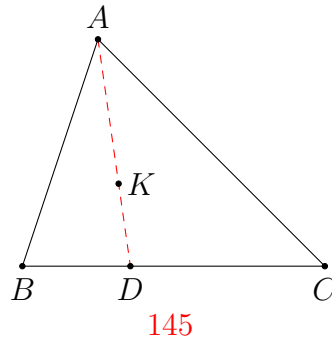


144. (4)(3) In $\triangle ABC$, let D be any point inside $\angle BAC$.

Then D lies on AK iff $\frac{d(D, AB)}{d(D, AC)} = \frac{AB}{AC}$.

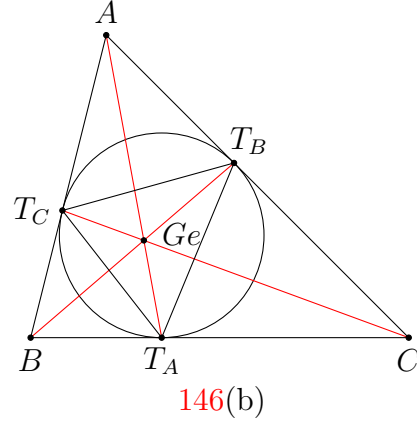
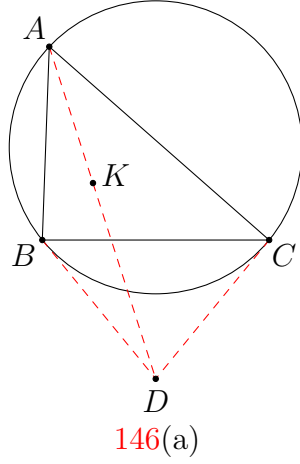
145. (4)(3) In $\triangle ABC$, let D be any point on the side BC .

Then D lies on AK iff $\frac{DB}{DC} = \frac{AB^2}{AC^2}$.

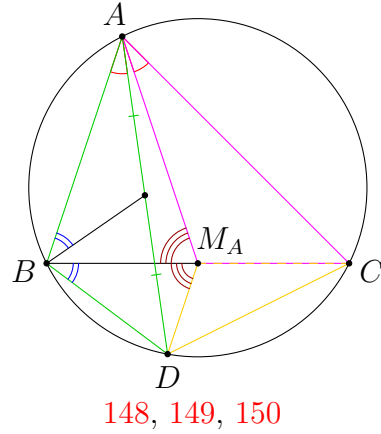
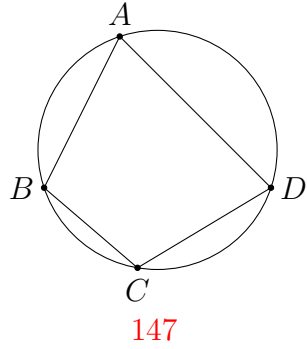


146. (5)(5) Let $\triangle ABC$ be a given triangle.
Then $\tan(B, \odot O) \cap \tan(C, \odot O) \cap AK \ (\sim)$.

(4)(1) So $Ge = K_{\triangle T_A T_B T_C}$.



147. (4)(4)(Def) Let $ABCD$ be a cyclic quadrilateral such that $\frac{AB}{AD} = \frac{CB}{CD}$.
Then $ABCD$ is called a *harmonic quadrilateral*(調和四邊形).

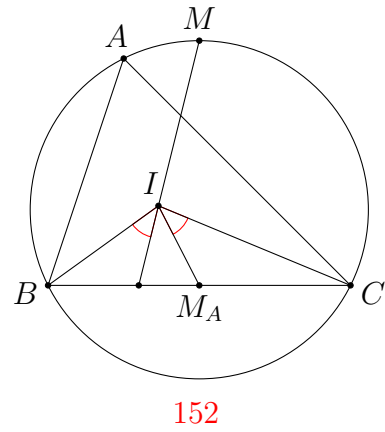
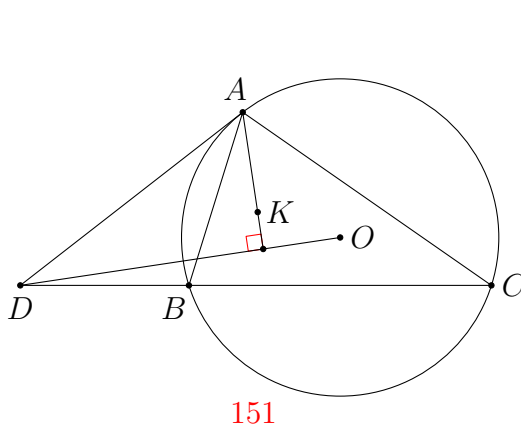


148. (4)(4) In $\triangle ABC$, let $D = AK \cap \odot O$.
Then $ABDC$ is a harmonic quadrilateral ($\sim$).

149. (4)(4) In $\triangle ABC$, let $D = AK \cap \odot O$.
Then AD is the D -symmedian of $\triangle DCB$ and BC is the B -symmedian of $\triangle BDA$ ($\sim$).

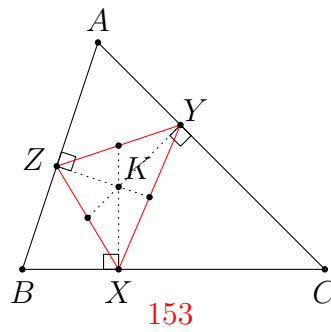
150. (4)(4) In $\triangle ABC$, let $D = AK \cap \odot O$.
 Then $\triangle AM_A C \sim \triangle ABD \sim \triangle CM_A D$ ($\sim$).
 (4)(3) So $\angle AM_A B = \angle DM_A B$ ($\sim$).

151. (4)(2) In $\triangle ABC$, let $D = BC \cap \tan(A, \odot O)$.
 Then $OD \perp AK$ ($\sim$).



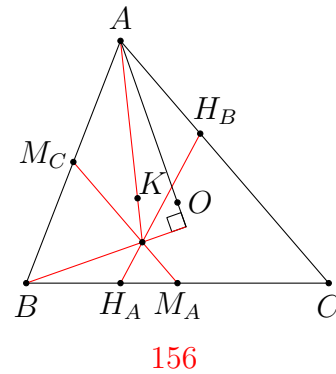
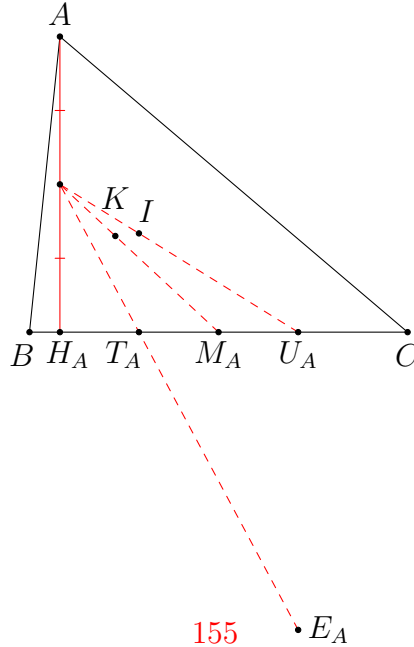
152. (3)(2) In $\triangle ABC$, let M be the midpoint of $\widehat{BAC}$ of $\odot O$.
 Then MI is the I -symmedian of $\triangle IBC$.

153. (2)(1) In $\triangle ABC$, let P be any interior point. Let $\triangle XYZ$ be the pedal triangle of P w.r.t. $\triangle ABC$.
 Then $P = G_{\triangle XYZ}$ iff $P = K$.



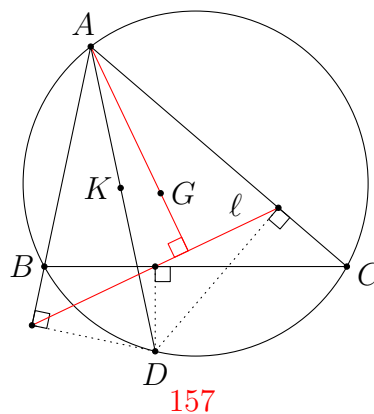
154. (2)(1) In $\triangle ABC$, let P be any interior point. Let $x = d(P, AB)$, $y = d(P, BC)$, $z = d(P, CA)$.
 Then $x^2 + y^2 + z^2$ has the smallest value when $P = K$.

155. (2)(1) Let $\triangle ABC$ be a given triangle.
Then $IU_A \cap EA T_A \cap M_A K = M_{AH_A}$ ($\sim$).



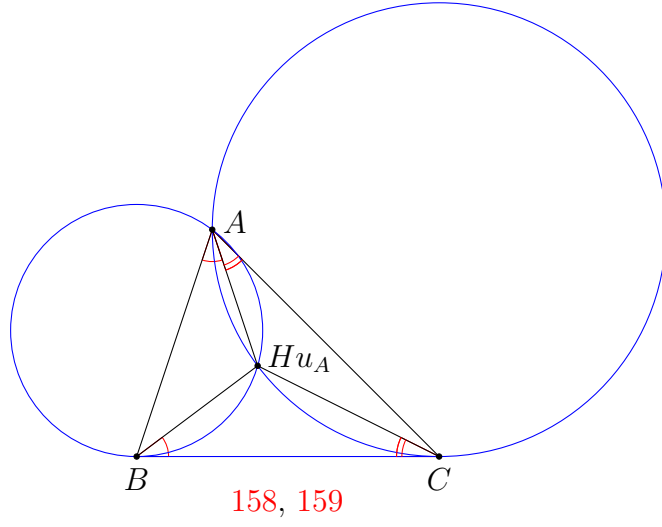
156. (1)(1) Let $\triangle ABC$ be a given triangle.
Then $AK \cap H_A H_B \cap M_A M_C \cap \perp (B, AO) (\sim)$.

157. (1)(1) In $\triangle ABC$, let $D = AK \cap \odot O$. Let ℓ be the Simson line of D w.r.t. $\triangle ABC$. Then $\ell \perp AG$ ($\sim$).

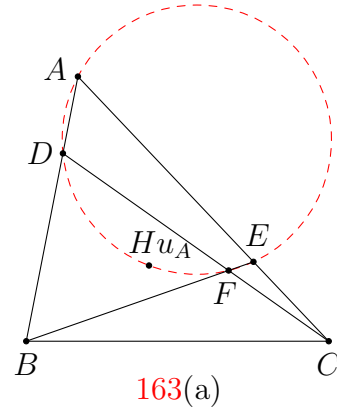
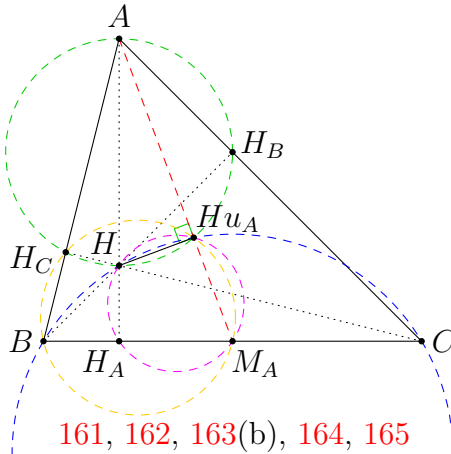


158. (4)(3)(Def) Let $\triangle ABC$ be an acute triangle.

Then the *A-humpty point* of $\triangle ABC$ is the point Hu_A in $\triangle ABC$ such that $\angle Hu_AAB = \angle Hu_ABC$ and $\angle Hu_AAC = \angle Hu_ACB$ (while in general, we need $\sphericalangle Hu_AAB = \sphericalangle Hu_ABC$ and $\sphericalangle Hu_AAC = \sphericalangle Hu_ACB$) ($\sim$).



159. (4)(2) Let $\triangle ABC$ be a given triangle.
Then BC is tangent to $(ABHu_A)$ and $(ACHu_A)$ ($\sim$).
160. (2)(1) Let $\triangle ABC$ be a given triangle.
Then $\frac{Hu_AB}{Hu_AC} = \frac{AB}{AC}$ ($\sim$).
161. (4)(3) Let $\triangle ABC$ be a given triangle.
Then A, Hu_A, M_A are collinear ($\sim$).
162. (4)(2) Let $\triangle ABC$ be a given triangle.
Then H, B, C, Hu_A are concyclic ($\sim$).



163. (2)(2) In $\triangle ABC$, let D, E be points on the sides AB, AC . Let $F = BE \cap CD$.

Then A, D, F, E are concyclic iff Hu_A lies on (ADE) ($\sim$).

(4)(2) So Hu_A lies on $(AH_C HH_B)$ and $\angle AHu_A H = 90^\circ$ ($\sim$).

164. (4)(2) Let $\triangle ABC$ be a given triangle.

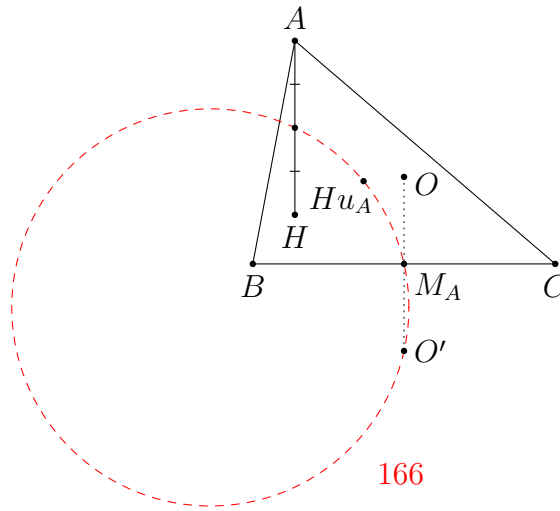
Then H, H_A, M_A, Hu_A are concyclic ($\sim$).

165. (3)(2) Let $\triangle ABC$ be a given triangle.

Then H_C, B, M_A, Hu_A are concyclic ($\sim$).

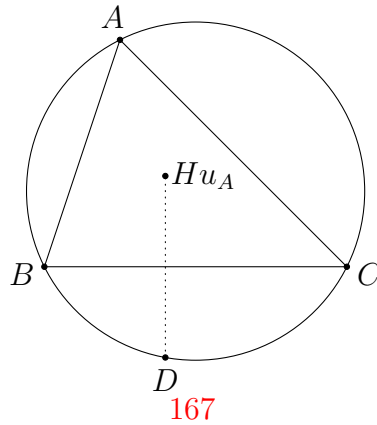
166. (1)(1) In $\triangle ABC$, let $O' = \text{Rf}(O, BC)$.

Then O', M_A, Hu_A, M_{AH} are concyclic ($\sim$).



167. (2)(1) In $\triangle ABC$, let $D = \text{Rf}(Hu_A, BC)$.

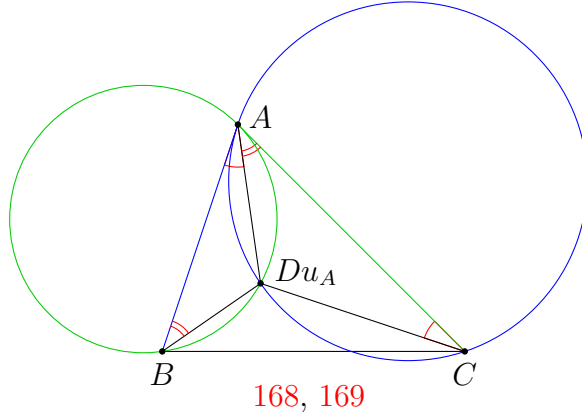
Then $ABDC$ is a harmonic quadrilateral ($\sim$).



168. (4)(3)(Def) Let $\triangle ABC$ be an acute triangle.

Then the A -*dummy point* of $\triangle ABC$ is the point Du_A in $\triangle ABC$ such that $\angle Du_A AB = \angle Du_A CA$ and $\angle Du_A AC = \angle Du_A BA$ (while in general, we need $\sphericalangle Du_A AB = \sphericalangle Du_A CA$ and $\sphericalangle Du_A AC = \sphericalangle Du_A BA$) ($\sim$).

(4)(3) So $\triangle Du_A BA \sim \triangle Du_A AC$ ($\sim$).



169. (4)(2) Let $\triangle ABC$ be a given triangle.

Then AB is tangent to $(ACDu_A)$ and AC is tangent to $(ABDu_A)$ ($\sim$).

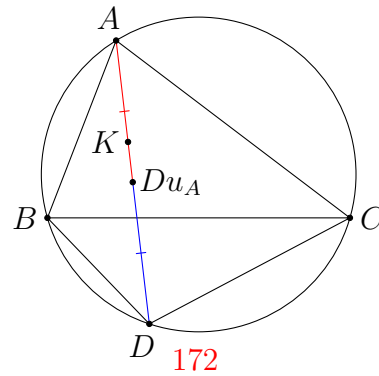
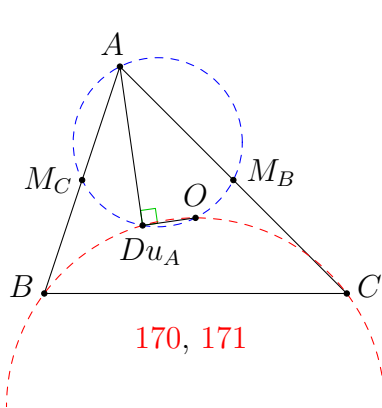
170. (4)(2) Let $\triangle ABC$ be a given triangle.

Then B, C, O, Du_A are concyclic ($\sim$).

171. (4)(2) Let $\triangle ABC$ be a given triangle.

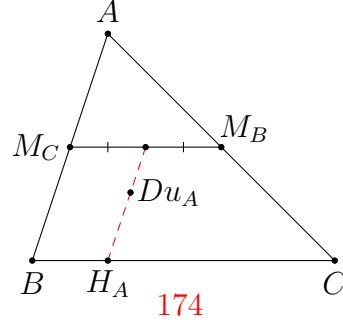
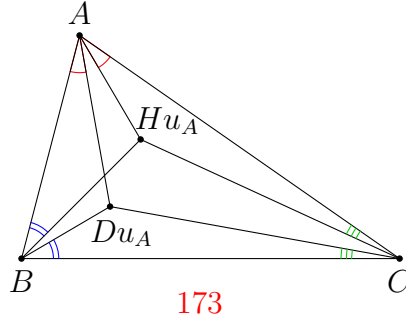
Then O, M_B, A, M_C, Du_A are concyclic ($\sim$).

(4)(2) So $\angle ADu_A O = 90^\circ$ ($\sim$).



172. (4)(3) In $\triangle ABC$, let $D = \text{Rf}(A, Du_A)$.
 Then $ABDC$ is a harmonic quadrilateral ($\sim$).
 (4)(3) So A, K, Du_A are collinear ($\sim$).

173. (4)(3) Let $\triangle ABC$ be a given triangle.
 Then $\text{Iso}(Hu_A) = Du_A$ ($\sim$).



174. (2)(1) Let $\triangle ABC$ be a given triangle.
 Then $H_A, Du_A, M_{M_B M_C}$ are collinear ($\sim$).

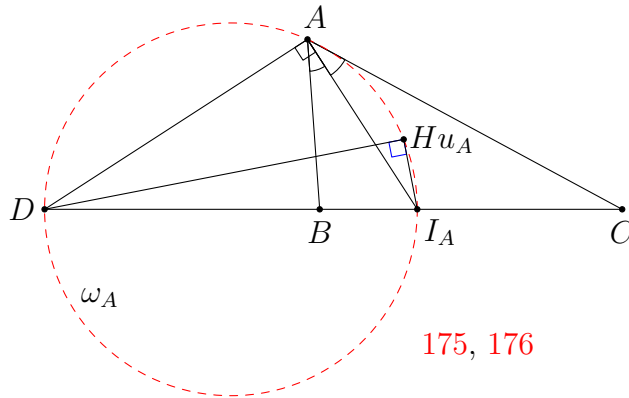
175. (5)(3)(Def) Let A and B be two distinct points and let $m : n$ be a fixed nonnegative ratio.

Then the locus of a point P such that $PA : PB = m : n$ is a circle, called an *Apollonius circle* (阿波羅尼斯圓), with centre lying on the line AB .

(5)(5) Limiting case: when $m = n$, the locus is the perpendicular bisector of AB .

(5)(2)(Def) In particular, let $\triangle ABC$ be a given triangle.

Then the circle ω_A of the locus of a point P such that $PB : PC = c : b$ is called the A -*Apollonius circle* (阿波羅尼斯圓) of $\triangle ABC$.

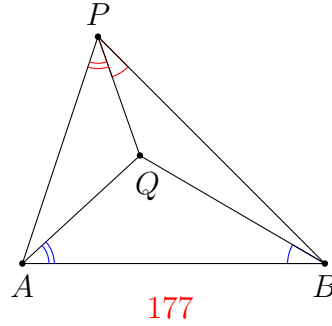


176. (3)(1) In $\triangle ABC$, let $D = BC \cap AE_B$.

Then A, D, I_A, Hu_A are concyclic, lying on ω_A ($\sim$).

177. (2)(2) Let A and B be fixed points. Let P and Q be two points such that $\frac{PA}{PB} = \frac{QA}{QB}$ and Q lies inside $\triangle PAB$.

Then $\angle APQ - \angle BPQ = \angle BAQ - \angle ABQ$ (while in general, we always have $\angle APQ + \angle BPQ = \angle BAQ + \angle ABQ$) ($\sim$).

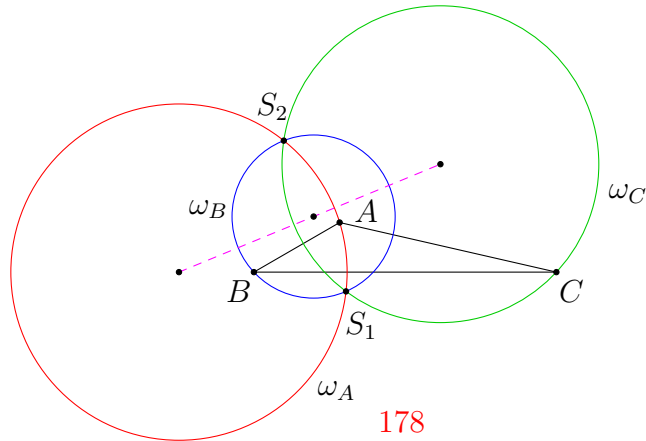


178. (2)(1)(Def) Let $\triangle ABC$ be a given triangle.

Then $\omega_A \cap \omega_B \cap \omega_C = S_1, S_2$, where S_1 lies inside $\odot O$, called the *1st, 2nd isodynamic points* (第一、第二等力點) of $\triangle ABC$.

(2)(1) So the centres of $\omega_A, \omega_B, \omega_C$ are collinear.

(2)(1) Limiting case: when $\triangle ABC$ is equilateral, S_2 is not defined.



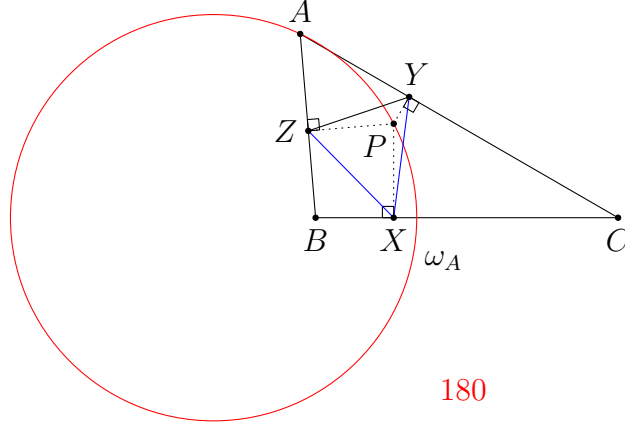
179. (2)(1) In $\triangle ABC$, let P be any point.

Then $AP : BP : CP = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}$ iff $P = S_1$ or S_2 .

180. (2)(1) In $\triangle ABC$, let $P \neq A, B, C$ be any point. Let $\triangle XYZ$ be the pedal triangle of P w.r.t. $\triangle ABC$.

Then $XY = XZ$ iff P lies on ω_A .

(2)(1) So $\triangle XYZ$ is an equilateral triangle iff $P = S_1$ or S_2 .

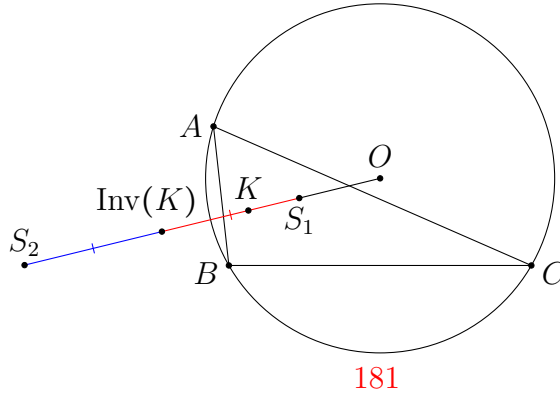


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181. (1)(1) Let $\triangle ABC$ be a non-equilateral triangle.

Then O, S_1, K, S_2 are collinear.

(1)(1) In particular, $\text{Inv}(S_1, \odot O) = S_2$ and $\text{Inv}(K, \odot O) = M_{S_1 S_2}$.



181

182. (3)(1)(Def) Let $\triangle ABC$ be a given triangle.

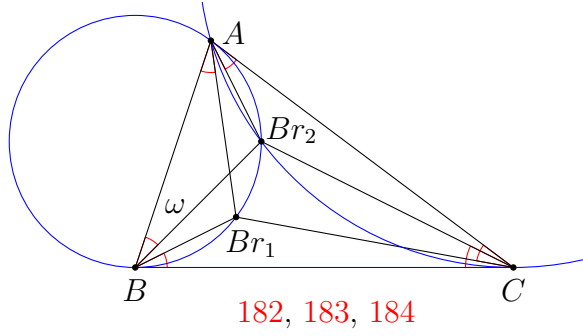
Then there is a unique point Br_1 inside $\triangle ABC$ such that

$$\angle BABr_1 = \angle CBBr_1 = \angle ACBr_1,$$

called the *1st Brocard point* (第一布羅卡點) of $\triangle ABC$; and there is a unique point Br_2 inside $\triangle ABC$ such that

$$\angle Br_2BA = \angle Br_2CB = \angle Br_2AC,$$

called the *2nd Brocard point* (第二布羅卡點) of $\triangle ABC$.



183. (3)(1) Let $\triangle ABC$ be a given triangle.

Then BC is tangent to $(ABBr_1)$ and $(ACBr_2)$ ($\sim$).

184. (3)(1) Let $\triangle ABC$ be a given triangle.

Then $\text{Iso}(Br_1) = Br_2$.

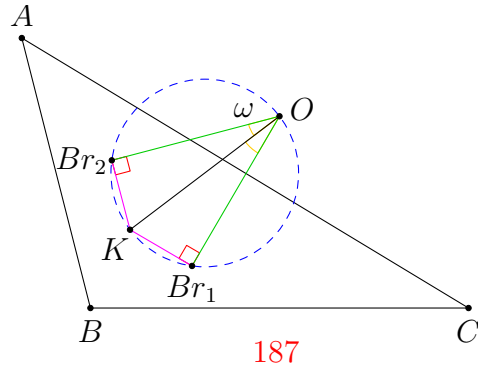
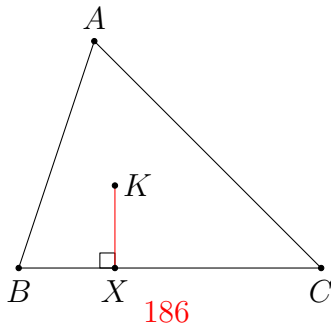
(3)(1)(Def) So $\angle BABr_1 = \angle Br_2BA = \omega$, called the *Brocard angle* (布羅卡角) of $\triangle ABC$ ($\sim$).

185. (3)(1) Let $\triangle ABC$ be a given triangle.

Then $\cot \omega = \cot A + \cot B + \cot C$.

186. (1)(1) In $\triangle ABC$, let $X = \perp K \cap BC$.

Then $KX = \frac{1}{2}a \tan \omega$ ($\sim$).



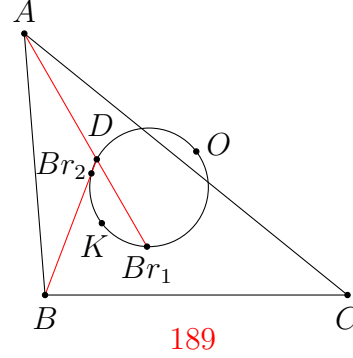
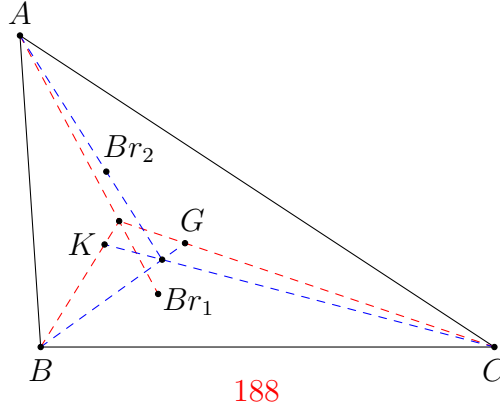
187. (1)(1) Let $\triangle ABC$ be a given triangle.

Then $\angle OBr_1K = \angle OBr_2K = 90^\circ$.

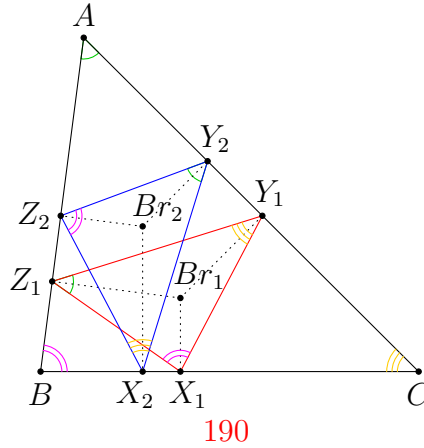
(1)(1) So O, Br_1, K, Br_2 are concyclic, lying on a circle with centre M_{OK} .

(1)(1) In particular, $\text{Rf}(Br_1, OK) = Br_2$ and $\angle Br_1OK = \angle Br_2OK = \omega$.

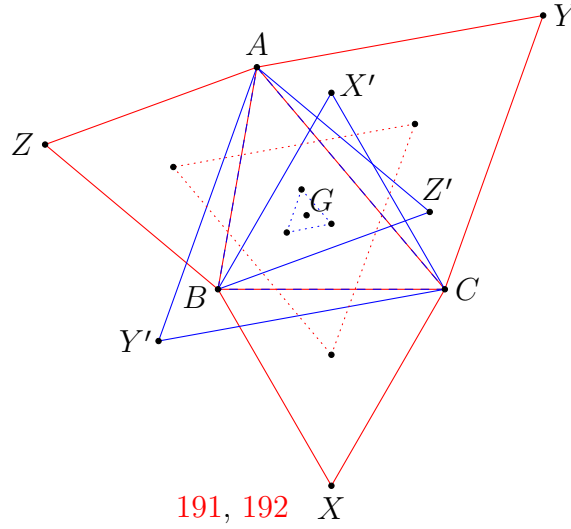
188. (1)(1) Let $\triangle ABC$ be a given triangle.
Then $ABr_1 \cap BK \cap CG$ and $ABr_2 \cap BG \cap CK$ ($\sim$).



189. (1)(1) In $\triangle ABC$, let $D = ABr_1 \cap BBr_2$.
Then D lies on (OBr_1KBr_2) ($\sim$).
190. (1)(1) In $\triangle ABC$, let $\triangle X_1Y_1Z_1$ and $\triangle X_2Y_2Z_2$ be the pedal triangles of Br_1 and Br_2 w.r.t. $\triangle ABC$ respectively.
Then $\triangle Z_1X_1Y_1 \sim \triangle Y_2Z_2X_2 \sim \triangle ABC$.



191. (4)(1)(Def) In $\triangle ABC$, construct equilateral triangles $\triangle CBX$, $\triangle ACY$, $\triangle BAZ$ such that X and A lie on different sides of BC , etc.
Then the centres of $\triangle CBX$, $\triangle ACY$, $\triangle BAZ$ form an equilateral triangle, called the *outer Napoleon triangle* (外拿破崙三角形) of $\triangle ABC$.
- (4)(1)(Def) Similarly, construct equilateral triangles $\triangle BCX'$, $\triangle CAY'$, $\triangle ABZ'$ such that X' and A lie on the same side of BC , etc.
Then the centres of $\triangle BCX'$, $\triangle CAY'$, $\triangle ABZ'$ form an equilateral triangle, called the *inner Napoleon triangle* (内拿破崙三角形) of $\triangle ABC$.



192. (4)(1) Let $\triangle ABC$ be a given triangle.

Then G is the centre of the outer Napoleon triangle of $\triangle ABC$, as well as the centre of the inner Napoleon triangle of $\triangle ABC$.

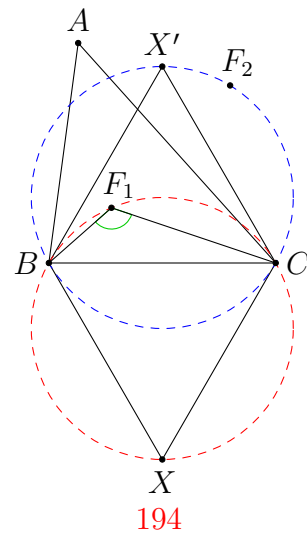
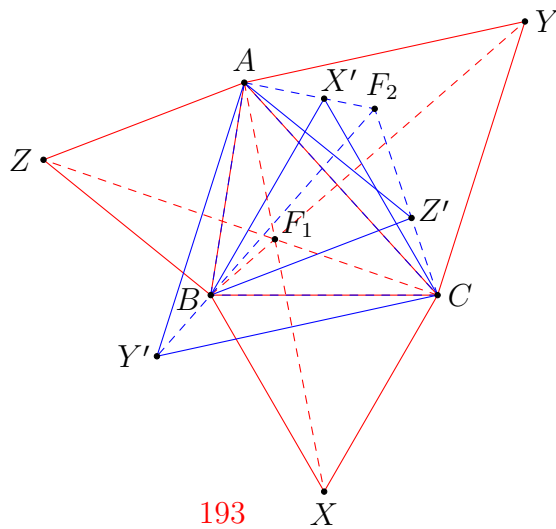
193. (3)(1)(Def) In $\triangle ABC$, let $\triangle CBX$, $\triangle ACY$, $\triangle BAZ$ be equilateral triangles constructed outwards.

Then $AX \cap BY \cap CZ = F_1$, called the *1st Fermat point* (第一費馬點) of $\triangle ABC$.

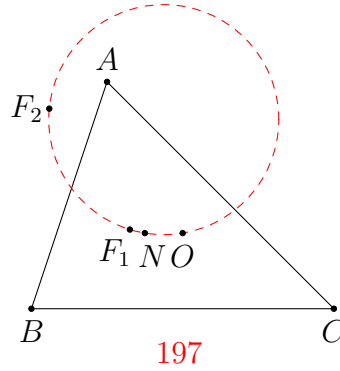
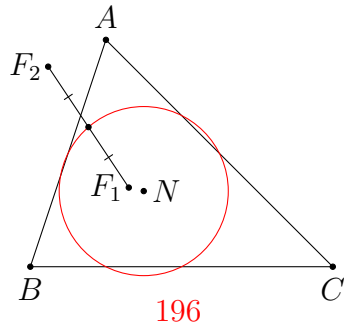
(3)(1)(Def) Similarly, let $\triangle BCX'$, $\triangle CAZ'$, $\triangle ABZ'$ be equilateral triangles constructed inwards.

Then $AX' \cap BY' \cap CZ' = F_2$, called the *2nd Fermat point* (第二費馬點) of $\triangle ABC$.

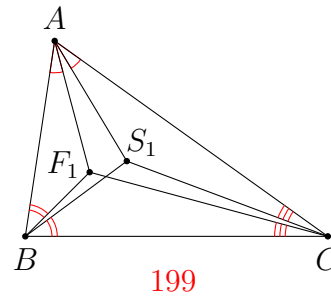
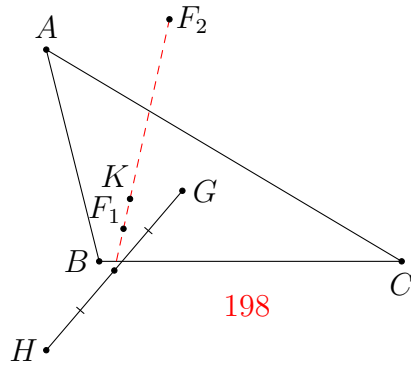
(3)(1) Limiting case: when $\triangle ABC$ is equilateral, F_2 is not defined.



194. (3)(1) In $\triangle ABC$, let $\triangle CBX$ be the equilateral triangle constructed outwards. Then B, X, C, F_1 are concyclic ($\sim$).
 (3)(1) In particular, if $A \leq 120^\circ$, then $\angle BF_1C = 120^\circ$ ($\sim$).
 (3)(1) Similarly, let $\triangle BCX'$ be the equilateral triangle constructed inwards. Then B, C, F_2, X' are concyclic ($\sim$).
195. (3)(1) In $\triangle ABC$ with $A, B, C \leq 120^\circ$, let P be any interior point. Then $PA + PB + PC$ has the smallest value when $P = F_1$.
196. (1)(1) Let $\triangle ABC$ be a non-equilateral triangle. Then $M_{F_1F_2}$ lies on $\odot N$.



197. (1)(1) Let $\triangle ABC$ be a non-equilateral triangle. Then O, N, F_1, F_2 are concyclic.
198. (1)(1) Let $\triangle ABC$ be a non-equilateral triangle. Then F_1, F_2, K, M_{GH} are collinear.



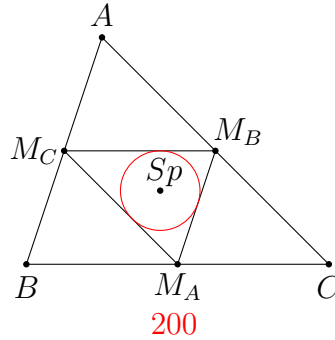
199. (1)(1) Let $\triangle ABC$ be a non-equilateral triangle.

Then $S_1 = \text{Iso}(F_1)$ and $S_2 = \text{Iso}(F_2)$.

200. (3)(1)(Def) Let $\triangle ABC$ be a given triangle.

Then $Sp = I_{\triangle M_A M_B M_C}$, called the *Spieker centre*(斯俾克圓心) of $\triangle ABC$.

(3)(1)(Def) In particular, $\odot \left(Sp, \frac{r}{2} \right) = \odot I_{\triangle M_A M_B M_C}$, called the *Spieker circle*(斯俾克圓) of $\triangle ABC$.



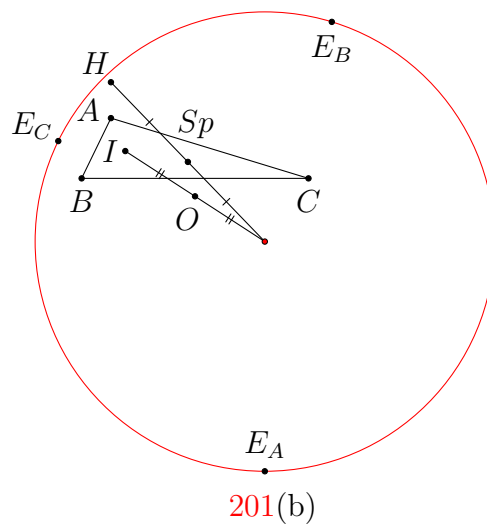
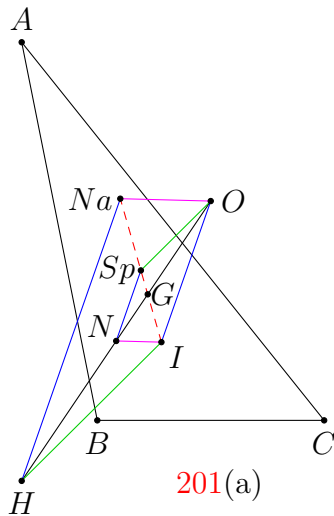
201. (3)(1) Let $\triangle ABC$ be a given triangle.

Then I, G, Sp, Na are collinear in that order and $IG : GSp : SpNa = 2 : 1 : 3$.

(3)(1)(Def) In particular, the line on which I, G, Sp, Na lie is called the *Nagel line*(奈格爾線) of $\triangle ABC$.

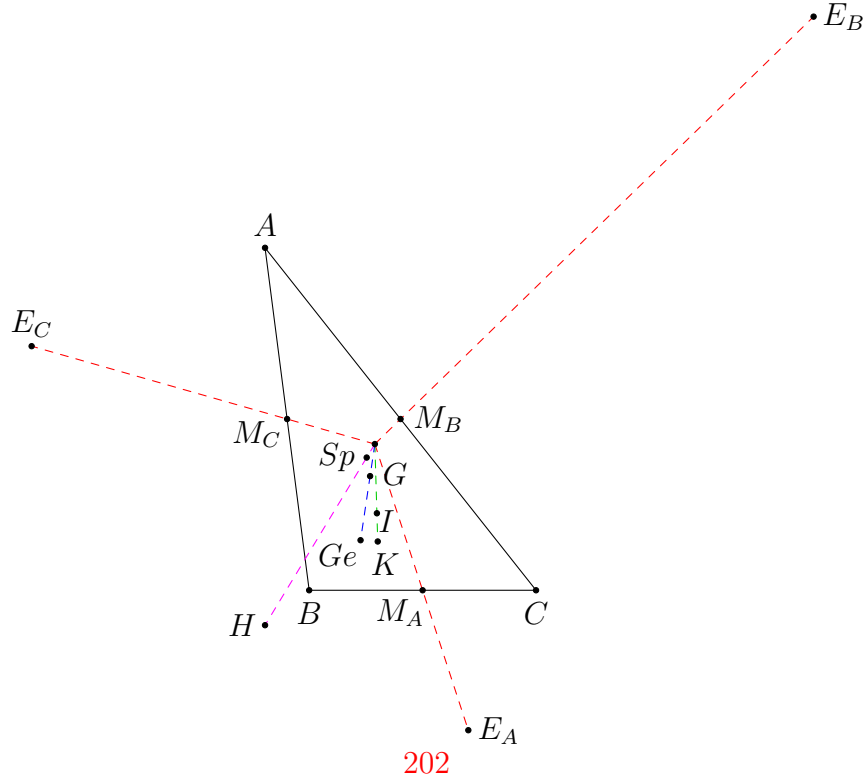
(2)(1) So $OI \parallel SpN \parallel NaH$, $OSp \parallel IH$ and $ONa \parallel IN$. Also, the centres of the internal and external homothety of $\odot I$ and $\odot I_{\triangle M_A M_B M_C}$ are G and Na respectively.

(1)(1) In particular, $\text{Rf}(H, Sp) = \text{Rf}(I, O) = O_{\triangle E_A E_B E_C}$.



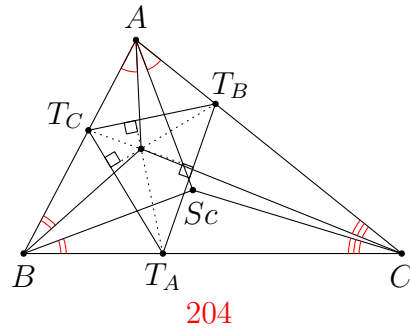
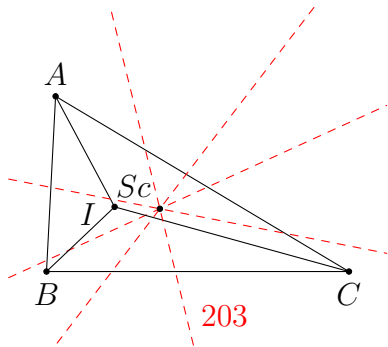
202. (1)(1) Let $\triangle ABC$ be a given triangle.

Then $E_A M_A \cap E_B M_B \cap E_C M_C \cap GGe \cap IK \cap HSp$.



203. (1)(1)(Def) Let $\triangle ABC$ be a given triangle.

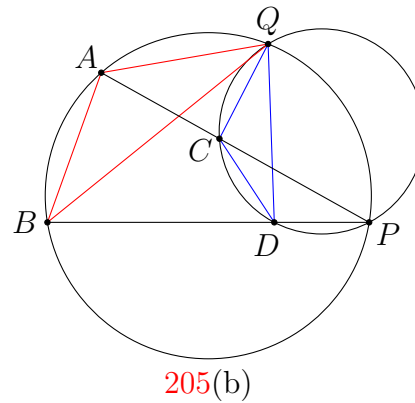
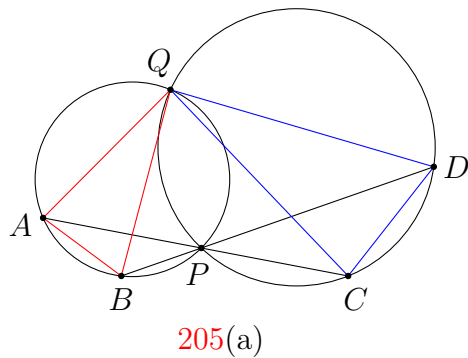
Then the Euler lines of $\triangle ABC$, $\triangle IBC$, $\triangle ICA$, $\triangle IAB$ are concurrent at the *Schiffler point* Sc of $\triangle ABC$.



204. (1)(1) Let $\triangle ABC$ be a given triangle.

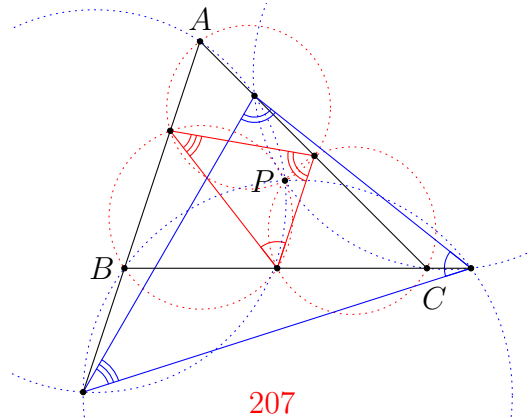
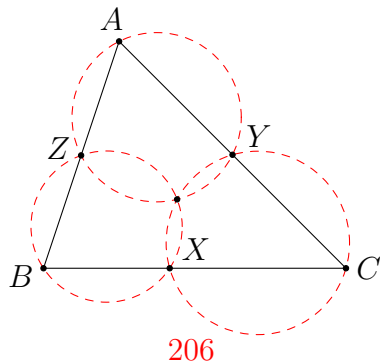
Then $\text{Iso}(Sc) = H_{\triangle T_A T_B T_C}$.

205. (5)(5) Let A, B, C, D be any points. Let $P = AC \cap BD$ and $Q = (ABP) \cap (CDP)$. Then Q is the centre of spiral similarity mapping AB to CD , as well as the centre of spiral similarity mapping AC to BD .



206. (5)(3) In $\triangle ABC$, let X, Y, Z be points on the lines BC, CA, AB respectively. Then $(AZY) \cap (BXZ) \cap (CYX)$.

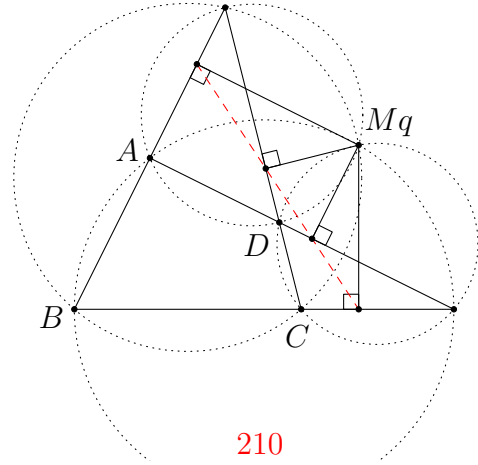
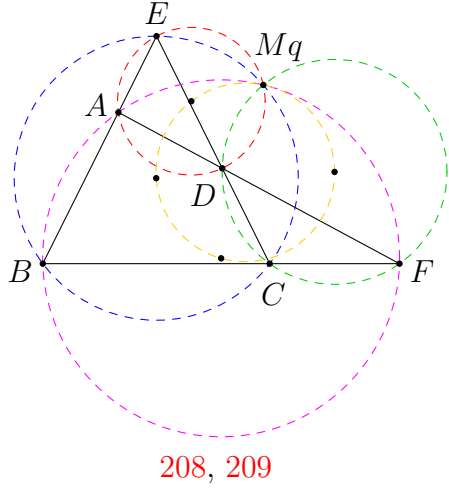
(5)(3)(Def) In particular, the intersection point is called the *Miquel point* (密克點) of $\triangle XYZ$ w.r.t. $\triangle ABC$.



207. (1)(1) In $\triangle ABC$, let P be any point not lying on the lines BC, CA, AB . Then all $\triangle XYZ$ which has P as the Miquel point w.r.t. $\triangle ABC$ are directly similar, with P being the centre of spiral similarity.
- (1)(1) In particular, every such $\triangle XYZ$ is directly similar to the pedal triangle of P w.r.t. $\triangle ABC$.
- (1)(1) So $\triangle XYZ$ is equilateral iff $P = S_1, S_2$.

208. (3)(3)(Def) In a quadrilateral $ABCD$, let $E = AB \cap CD$ and $F = AD \cap BC$.
Then $(ADE) \cap (BCE) \cap (CDF) \cap (ABF) = Mq$, called the *Miquel point*(密克點) of $ABCD$.

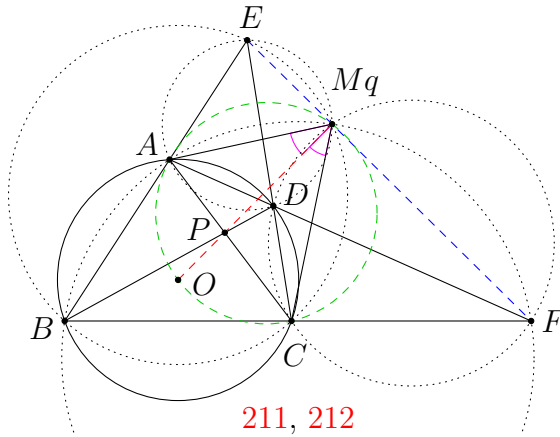
209. (2)(1) In a quadrilateral $ABCD$, let $E = AB \cap CD$ and $F = AD \cap BC$.
Then $O_{\triangle ADE}, O_{\triangle BCE}, O_{\triangle CDF}, O_{\triangle ABF}, Mq$ are concyclic.



210. (2)(1) Let $ABCD$ be a given quadrilateral.
Then $\perp Mq \cap AB, \perp Mq \cap BC, \perp Mq \cap CD, \perp Mq \cap DA$ are collinear.

211. (3)(3) In a cyclic quadrilateral $ABCD$, let $P = AC \cap BD$, $E = AB \cap CD$ and $F = AD \cap BC$.
Then $Mq = \text{Inv}(P, \odot O)$.

(3)(3) So O, P, Mq are collinear, and E, Mq, F are collinear.



212. (3)(2) Let $ABCD$ be a cyclic quadrilateral.

Then A, O, C, M_q are concyclic ($\sim$).

(3)(1) So $\angle AM_qO = \angle CM_qO$ ($\sim$).

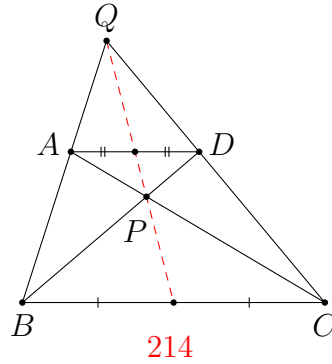
213. (5)(5)(Def) Let $ABCD$ be a convex quadrilateral with $AD \parallel BC$.

Then $ABCD$ is called a *trapezoid* (梯形).

(5)(5)(Def) In particular, when $AB = CD$ and $\angle ABC = \angle BCD$, then $ABCD$ is called an *isosceles trapezoid* (等腰梯形).

214. (4)(3) (Steiner's theorem) Let $ABCD$ be a trapezoid such that $AD \parallel BC$. Let $P = AC \cap BD$ and $Q = AB \cap CD$.

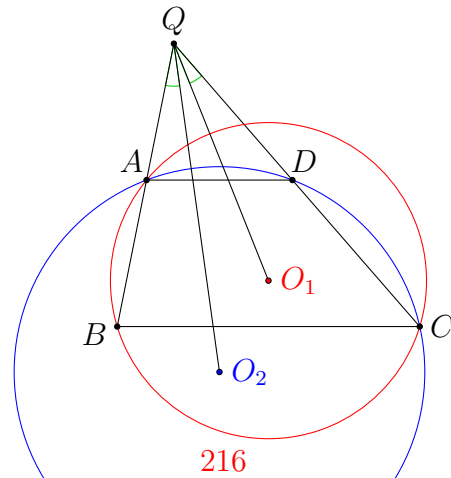
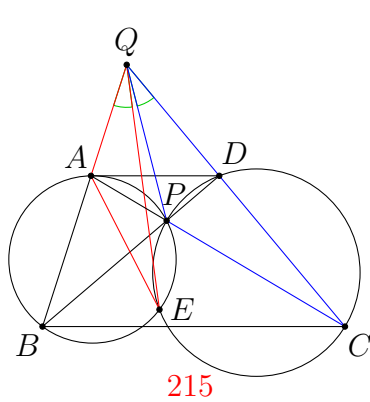
Then M_{BC}, P, M_{AD}, Q are collinear.



215. (2)(2) Let $ABCD$ be a trapezoid such that $AD \parallel BC$. Let $P = AC \cap BD$, $Q = AB \cap CD$ and $E = (ABP) \cap (CDP)$.

Then $\triangle QAE \sim \triangle QPC$ ($\sim$).

(2)(2) (Balkan MO 2009 Problem 2) So $\angle AQE = \angle PQC$ ($\sim$).



216. (1)(1) Let $ABCD$ be a trapezoid such that $AD \parallel BC$. Let $Q = AB \cap CD$, $O_1 = O_{\triangle ABC}$ and $O_2 = O_{\triangle ACD}$.

Then $\angle BQO_2 = \angle O_1QC$ ($\sim$).

217. (5)(5)(Def) Let $ABCD$ be a convex quadrilateral with $AB \parallel DC$ and $AD \parallel BC$.

Then $ABCD$ is called a *parallelogram*(平行四邊形).

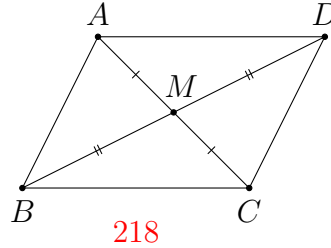
(5)(5) In particular, $ABCD$ is a trapezoid.

218. (5)(5) Let $ABCD$ be a convex quadrilateral.

Then $ABCD$ is a parallelogram iff $M_{AC} = M_{BD}$.

(5)(1) So let $M = M_{AC} = M_{BD}$.

Then $[ADM] = \frac{1}{4}[ABCD]$ ($\sim$).

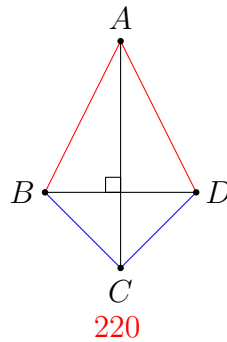


219. (5)(4)(Def) Let $ABCD$ be a convex quadrilateral with $a = d$ and $b = c$.

Then $ABCD$ is called a *kite*(鷓形).

220. (5)(4) Let $ABCD$ be a kite.

Then $AC \perp BD$.

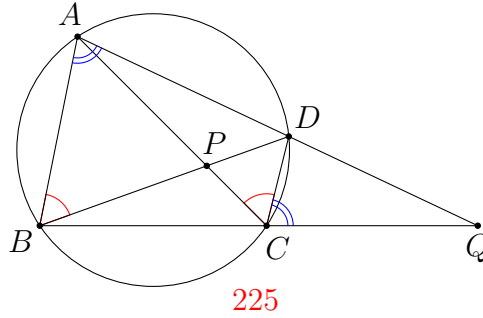


221. (5)(3)(Def) Let $ABCD$ be a convex quadrilateral with $a = b = c = d$.

Then $ABCD$ is called a *rhombus*(菱形).

(5)(3) In particular, $ABCD$ is a parallelogram and a kite.

222. (5)(3)(Def) Let $ABCD$ be a convex quadrilateral with $A = B = C = D = 90^\circ$.
Then $ABCD$ is called a *rectangle*(長方形).
(5)(3) In particular, $ABCD$ is a parallelogram.
223. (5)(3)(Def) Let $ABCD$ be a convex quadrilateral with $a = b = c = d$ and $A = B = C = D = 90^\circ$.
Then $ABCD$ is called a *square*(正方形).
(5)(3) In particular, $ABCD$ is a rectangle and a rhombus.
224. (5)(5)(Def) Let $ABCD$ be a convex quadrilateral with vertices lying on a circle.
Then $ABCD$ is called a *cyclic quadrilateral*, or an *inscribed quadrilateral*(圓內接四邊形), while points A, B, C, D are called *concylic*(共圓).
225. (5)(5) In a convex quadrilateral $ABCD$, let $P = AC \cap BD$ and $Q = AD \cap BC$.
Then $ABCD$ is cyclic iff one of the following holds:
- (i) $\angle ABD = \angle ACD$ ($\sim$);
 - (ii) $A + C = 180^\circ$ ($\sim$);
 - (iii) $A = \angle DCQ$ ($\sim$);
 - (iv) $PA \cdot PC = PB \cdot PD$;
 - (v) $QA \cdot QD = QB \cdot QC$.



226. (5)(1) In a convex quadrilateral $ABCD$, let $p = AC$, $q = BD$, and let θ be the angle between the lines AC and BD .
Then $[ABCD] = \frac{1}{2}pq \sin \theta$.
227. (2)(1) In a convex quadrilateral $ABCD$, let θ be the angle between the lines AC and BD .
Then $[ABCD] = \frac{1}{4} |(b^2 + d^2 - a^2 - c^2) \tan \theta|$.

228. (2)(1) (Bretschneider's formula(布雷特斯奈德公式)) Let $ABCD$ be a convex quadrilateral.

$$\text{Then } [ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd \cos^2 \frac{A+C}{2}}.$$

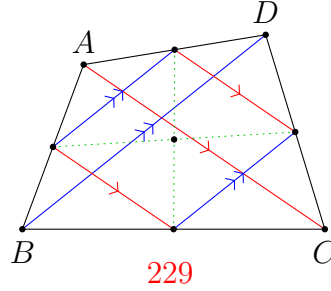
(3)(1) (Brahmagupta's formula(婆羅摩笈多公式)) In particular, when A, B, C, D are concyclic, $[ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.

(5)(3) (Heron's formula(海倫公式)) Limiting case: when D and A coincide, $[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$.

229. (5)(2) Let $ABCD$ be a convex quadrilateral.

Then $M_{AB}M_{BC} \parallel M_{DA}M_{CD} \parallel AC$ ($\sim$).

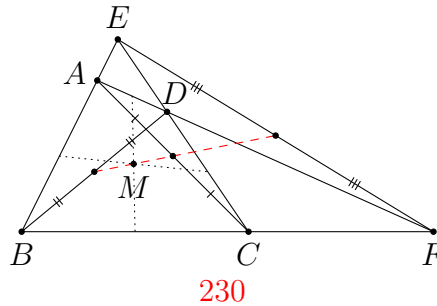
(4)(2) So $M_{AB}M_{BC}M_{CD}M_{DA}$ is a parallelogram and $M_{M_{AB}M_{CD}} = M_{M_{BC}M_{DA}}$.



230. (4)(1) In a convex quadrilateral $ABCD$, let $E = AB \cap CD$, $F = AD \cap BC$ and $M = M_{AB}M_{CD} \cap M_{BC}M_{DA}$.

Then $M_{AC}, M_{BD}, M_{EF}, M$ are collinear.

(4)(1)(Def) In particular, the line on which $M_{AC}, M_{BD}, M_{EF}, M$ lie is called the *Newton-Gauss line*(牛頓-高斯線) of quadrilateral $ABCD$.

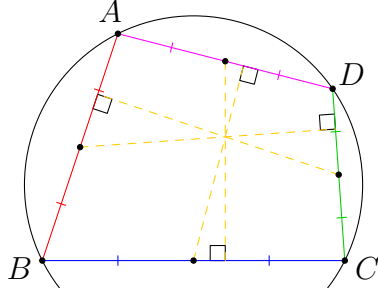


231. (1)(1) Let $ABCD$ be a cyclic quadrilateral.

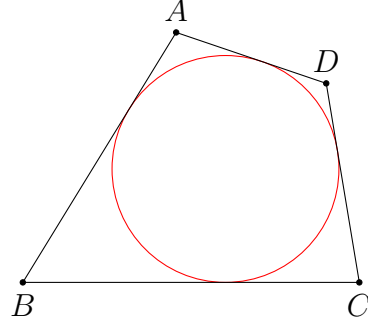
$$\text{Then } R = \frac{1}{4} \sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{(s-a)(s-b)(s-c)(s-d)}}.$$

232. (2)(1) Let $ABCD$ be a cyclic quadrilateral.

Then $\perp (M_{AB}, CD) \cap \perp (M_{BC}, DA) \cap \perp (M_{CD}, AB) \cap \perp (M_{DA}, BC)$.



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233, 234(a)

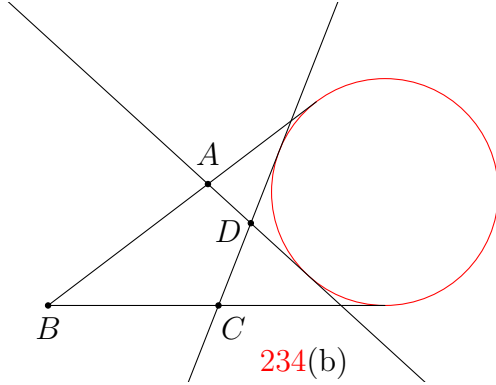
233. (5)(3)(Def) Let $ABCD$ be a convex quadrilateral with an inscribed circle.

Then $ABCD$ is called a *circumscribed quadrilateral* (圓外接四邊形).

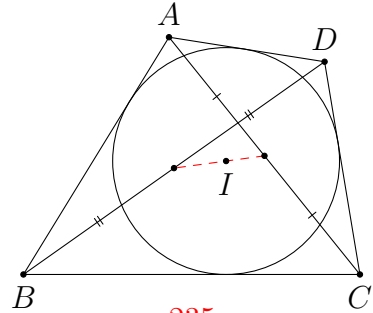
234. (5)(3) (Pitot theorem (皮托定理)) Let $ABCD$ be a convex quadrilateral.

Then $ABCD$ is a circumscribed quadrilateral iff $a + c = b + d$.

(4)(2) Similarly, $ABCD$ has an exscribed circle opposite to B or D iff $b + c = d + a$ ($\sim$).



234(b)



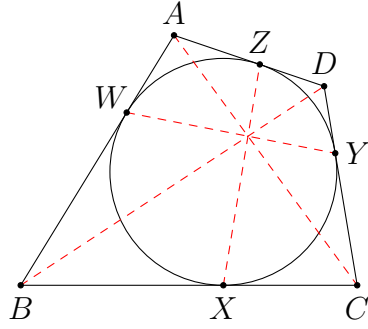
235

235. (4)(1) (Newton's theorem (牛頓定理)) Let $ABCD$ be a circumscribed quadrilateral.

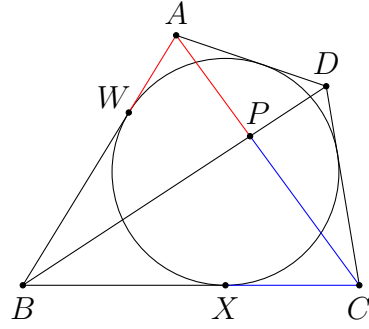
Then I lies on the Newton-Gauss line of $ABCD$.

236. (4)(2) Let $ABCD$ be a circumscribed quadrilateral. Let $W = \odot I \cap AB$, $X = \odot I \cap BC$, $Y = \odot I \cap CD$, $Z = \odot I \cap DA$.

Then $AC \cap BD \cap WY \cap XZ$.



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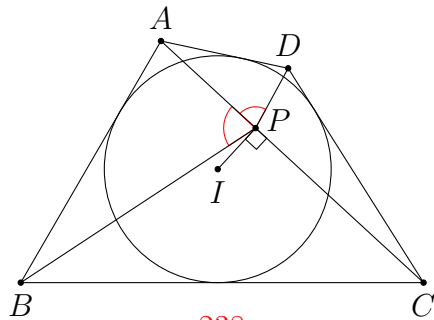
237

237. (1)(1) Let $ABCD$ be a circumscribed quadrilateral. Let $P = AC \cap BD$, $W = \odot I \cap AB$, $X = \odot I \cap BC$.

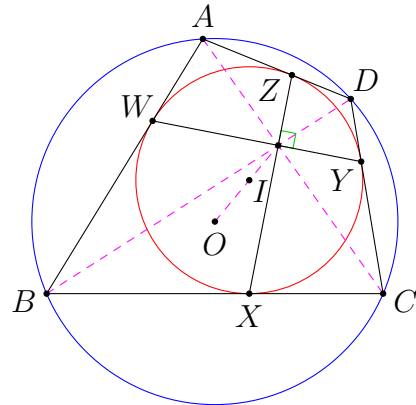
Then $\frac{AP}{AW} = \frac{CP}{CX} (\sim)$.

238. (1)(1) Let $ABCD$ be a circumscribed quadrilateral. Let $P = \perp I \cap AC$.

Then $\angle APB = \angle APD (\sim)$.



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239, 240, 241

239. (3)(1)(Def) Let $ABCD$ be a cyclic and circumscribed quadrilateral.

Then $ABCD$ is called a *bicentric quadrilateral* (雙心四邊形).

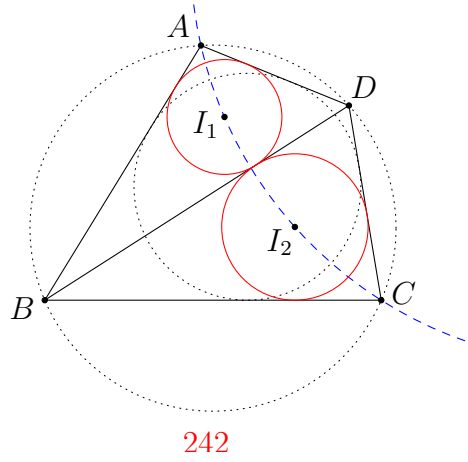
240. (3)(1) Let $ABCD$ be a bicentric quadrilateral. Let $W = \odot I \cap AB$, $X = \odot I \cap BC$, $Y = \odot I \cap CD$, $Z = \odot I \cap DA$.

Then $WY \perp XZ$.

241. (3)(1) Let $ABCD$ be a bicentric quadrilateral.

Then $AC \cap BD \cap OI$.

242. (1)(1) Let $ABCD$ be a bicentric quadrilateral. Let $I_1 = I_{\triangle ABD}$, $I_2 = I_{\triangle BCD}$.
Then A, I_1, I_2, C are concyclic ($\sim$).



243. (1)(1) Let $ABCD$ be a bicentric quadrilateral.

Then $r = \frac{\sqrt{abcd}}{s}$ and $\frac{1}{(R - OI)^2} + \frac{1}{(R + OI)^2} = \frac{1}{r^2}$.

244. (5)(4)(Def) In a circle Γ with centre O , let AB be a chord.

Then AB has the longest length when A, O, B are collinear, where the segment AB is called a *diameter* (直徑) of Γ .

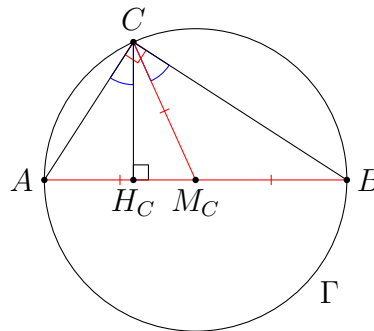
245. (5)(5) Let A, B, C be any points on a circle Γ with centre O .

Then $\angle AOB = 2\angle ACB$ (if $\angle ACB \leq 90^\circ$) or $\angle AOB = 360^\circ - 2\angle ACB$ (if $\angle ACB > 90^\circ$).

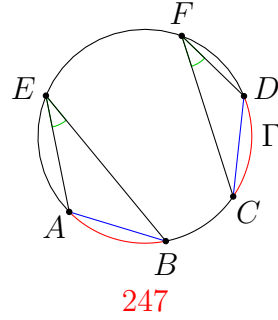
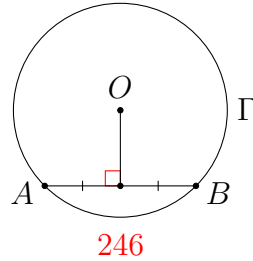
(5)(5) In particular, AB is a diameter of Γ iff $\angle ACB = 90^\circ$.

(5)(5) So let $\triangle ABC$ be a given triangle.

Then $C = 90^\circ$ iff $CM_C = AM_C = BM_C$ iff $\angle ACH_C = \angle BCM_C$.



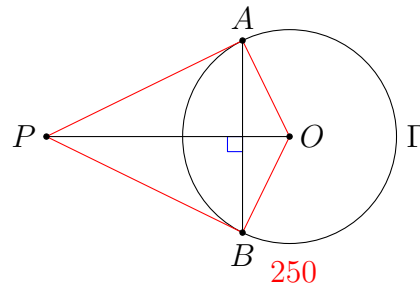
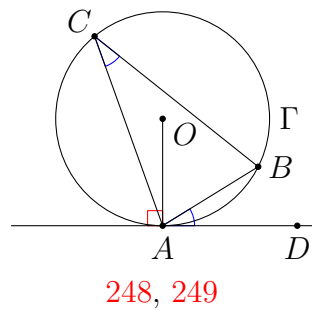
246. (5)(5) Let AB be a chord of a circle Γ with centre O which is not a diameter.
Then $OM_{AB} \perp AB$.



247. (5)(4) Let AB and CD be chords of a circle Γ . Let E and F be points on the major (or minor) $\widehat{AB}$ and $\widehat{CD}$ respectively.
Then the following conditions are equivalent:
(i) $\widehat{AB} = \widehat{CD}$;
(ii) $AB = CD$;
(iii) $\angle AEB = \angle CFD$.

248. (5)(5) Let A be any point on a circle Γ with centre O .
Then $\tan(A, \Gamma) \perp OA$.

249. (5)(5) Let A, B, C be any points on a circle Γ . Let D be a point on $\tan(A, \Gamma)$ such that B and D lie on the same side of AC .
Then $\angle ACB = \angle DAB$ ($\sim$).



250. (5)(5) Let AB be a chord of a circle Γ with centre O which is not a diameter.
Let $P = \tan(A, \Gamma) \cap \tan(B, \Gamma)$.
Then $\triangle PAO \cong \triangle PBO$.
(5)(5) So $PAOB$ is a kite and $PO \perp AB$.

251. (5)(5)(Def) Let $\Gamma = \odot(O, r)$ be a given circle and let P be any point.

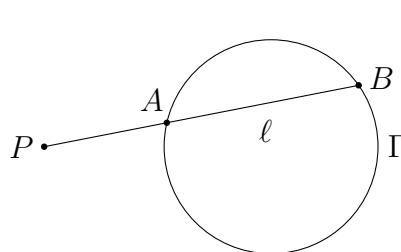
Then $\tilde{P}_\Gamma = OP^2 - r^2$, called the *power*(幂) of P w.r.t. Γ .

(5)(5) So $\tilde{P}_\Gamma = 0$ iff P lies on Γ , $\tilde{P}_\Gamma > 0$ iff P lies outside Γ , $\tilde{P}_\Gamma < 0$ iff P lies inside Γ .

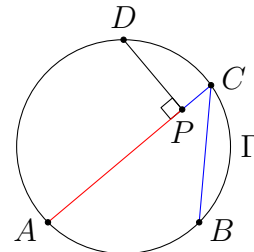
252. (5)(5) (Power of a point theorem(圓幂定理)) Let Γ be a given circle and let P be any point. Let $A, B = \ell(P) \cap \Gamma$.

Then $PA \cdot PB = \tilde{P}_\Gamma$ (expressed in directed lengths).

(5)(5) Limiting case: when $\ell(P)$ is a tangent to Γ , $A = B$ and $PA^2 = \tilde{P}_\Gamma$.



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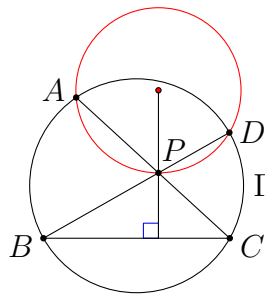
253

253. (3)(1) (Archimedes' broken-chord theorem(阿基米德中點定理)) Let A, B, C be any points on a circle Γ with $AC > BC$. Let D be the midpoint of $\widehat{ACB}$, and let $P = \perp D \cap AC$.

Then $AP = BC + CP$.

254. (2)(1) Let A, B, C, D be any points on a circle Γ . Let $P = AC \cap BD$.

Then $BC \perp PO_{\triangle APD}$ ($\sim$).



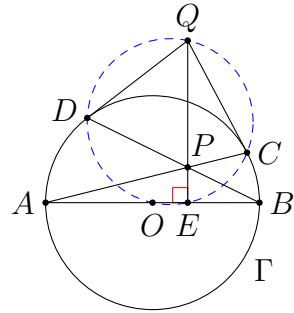
254

255. (2)(2) (IMO Shortlist 1966) Let A, B, C, D be any points on a circle Γ with centre O such that AB is a diameter. Let $P = AC \cap BD$ and $Q = \tan(C, \Gamma) \cap \tan(D, \Gamma)$.

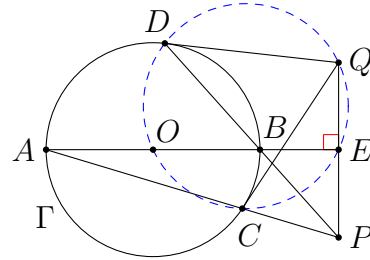
Then $PQ \perp AB$.

(2)(1) (CWMO 2010 Problem 2) So let $E = PQ \cap AB$.

Then C, Q, D, O, E are concyclic.



255(a)

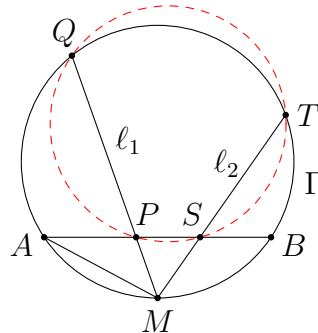


255(b)

256. (4)(3) (Shooting lemma) Let AB be a chord of a circle Γ , and let M be the midpoint of $\widehat{AB}$. Let $P, Q = \ell_1(M) \cap AB, \Gamma$ and $S, T = \ell_2(M) \cap AB, \Gamma$.

Then $MP \times MQ = MA^2$ ($\sim$).

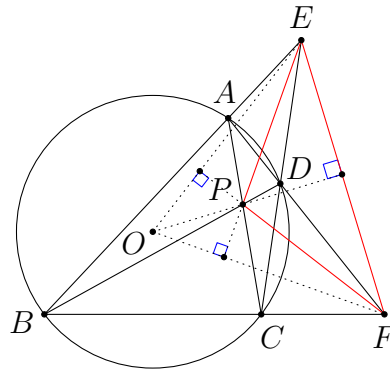
(4)(3) So Q, P, S, T are concyclic.



256

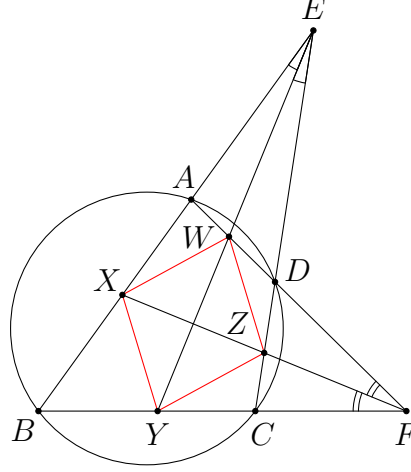
257. (5)(3) (Brokard's theorem(勃羅卡定理)) In a cyclic quadrilateral $ABCD$, let $P = AC \cap BD$, $E = AB \cap CD$ and $F = AD \cap BC$.

Then $O = H_{\triangle EFP}$ ($\sim$).



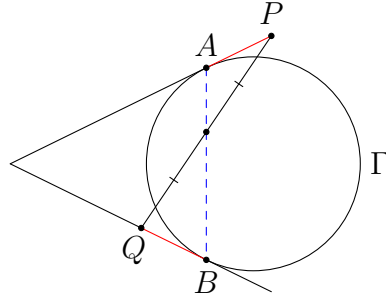
257

258. (1)(1) In a cyclic quadrilateral $ABCD$, Let $E = AB \cap CD$ and $F = AD \cap BC$. Let ℓ_1 and ℓ_2 be the internal angle bisectors of $\angle BEC$ and $\angle AFB$, and let $W, Y = \ell_1 \cap AD, BC$ and $X, Z = \ell_2 \cap AB, CD$. Then $WXYZ$ is a rhombus.

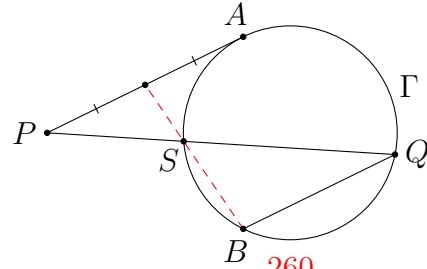


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259. (1)(1) Let A and B be any points on a circle Γ . Let P and Q be points on $\tan(A, \Gamma)$ and $\tan(B, \Gamma)$ respectively such that they lie on different sides of AB . Then $AP = BQ$ iff A, M_{PQ}, B are collinear.



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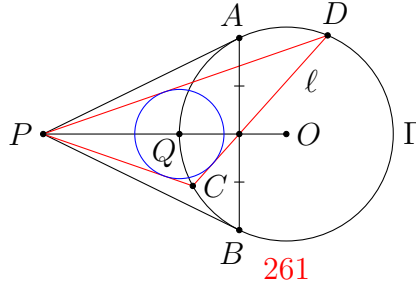
260

260. (2)(1) (All-Soviet Union MO 1973 Problem 177) Let A and B be any points on a circle Γ . Let $P = \tan(A, \Gamma) \cap \tan(B, \Gamma)$, $Q = \ell(B, PA) \cap \Gamma$ and $S = PQ \cap \Gamma$. Then B, S, M_{PA} are collinear ($\sim$).

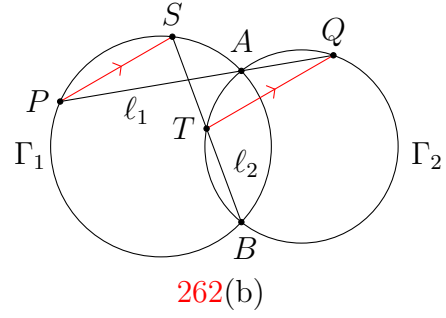
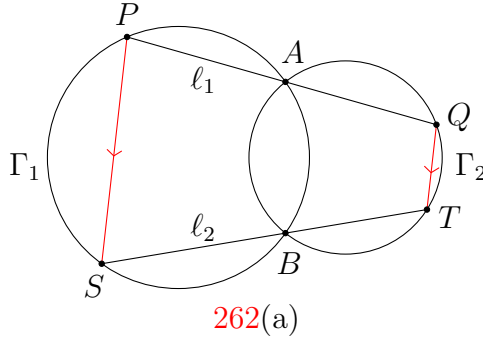
261. (1)(1) (CWMO 2001 Problem 6) Let A and B be any points on a circle Γ with centre O . Let $P = \tan(A, \Gamma) \cap \tan(B, \Gamma)$, $Q = PO \cap \Gamma$ (segment PO) and $C, D = \ell(M_{AB}) \cap \Gamma$.

Then $Q = I_{\triangle PCD}$.

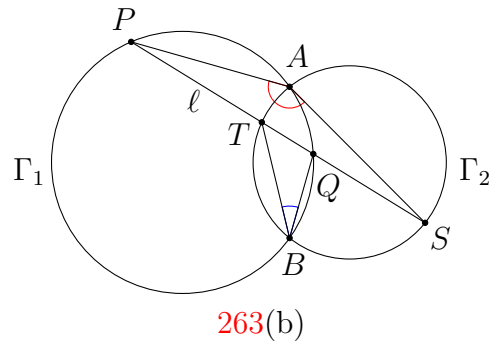
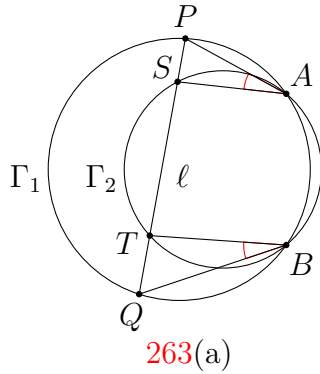
(3)(2) In particular, when $\ell = AB$, $Q = I_{\triangle PAB}$.



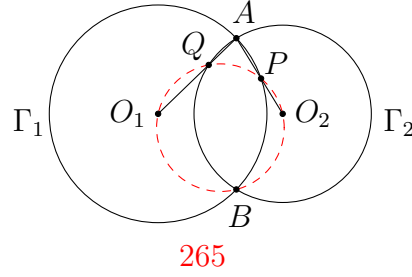
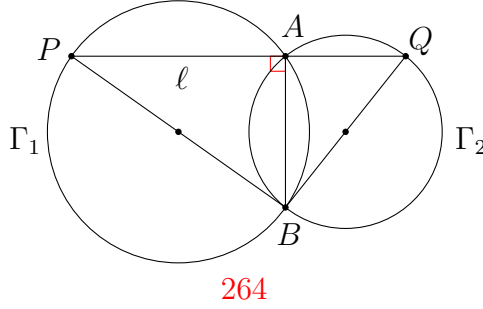
262. (5)(5) (Reim's theorem(雷姆定理)) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let $P, Q = \ell_1(A) \cap \Gamma_1, \Gamma_2$ and $S, T = \ell_2(B) \cap \Gamma_1, \Gamma_2$. Then $PS \parallel QT$.



263. (4)(3) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let $P, Q = \ell \cap \Gamma_1$, $S, T = \ell \cap \Gamma_2$. Then $\angle SAP = \angle QBT$ or $\angle SAP + \angle QBT = 180^\circ$ (while $\angle SAP = \angle QBT$ always holds).
 (4)(3) Limiting case: when ℓ is tangent to Γ_2 , the same holds with $S = T$ ($\sim$); when Γ_1 is tangent to Γ_2 , the same holds with $A = B$.

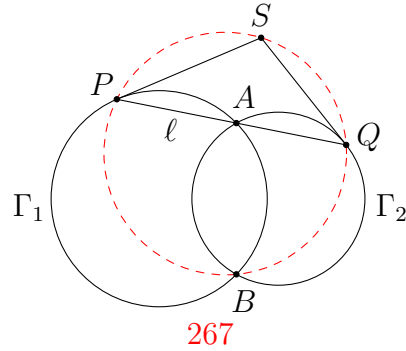
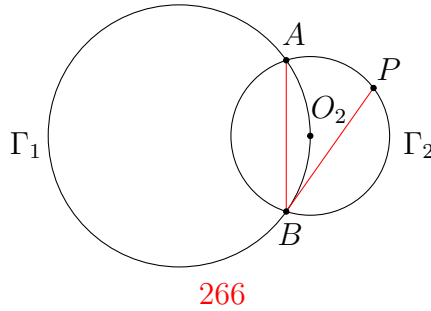


264. (1)(1) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let $P, Q = \ell(A) \cap \Gamma_1, \Gamma_2$. Then PQ has the longest length when $\ell \perp AB$, i.e. BP, BQ are diameters of Γ_1, Γ_2 respectively ($\sim$).



265. (1)(1) Let Γ_1, Γ_2 be two circles with centres O_1, O_2 respectively which intersect at A, B . Let $P = O_2A \cap \Gamma_1$, $Q = O_1A \cap \Gamma_2$. Then O_1, B, O_2, P, Q are concyclic ($\sim$).

266. (1)(1) Let Γ_1, Γ_2 be two circles with centres O_1, O_2 respectively which intersect at A, B . Let $P = \tan(B, \Gamma_1) \cap \Gamma_2$. Then O_2 lies on Γ_1 iff $BA = BP$ ($\sim$).



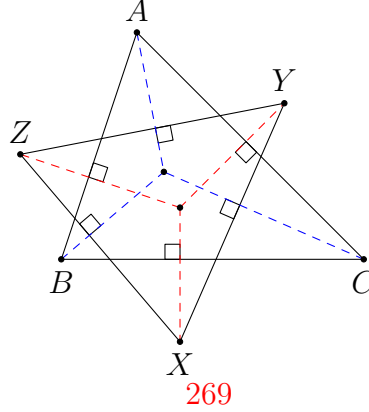
267. (1)(2) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let $P, Q = \ell(A) \cap \Gamma_1, \Gamma_2$ and $S = \tan(P, \Gamma_1) \cap \tan(Q, \Gamma_2)$. Then P, B, Q, S are concyclic ($\sim$).

268. (5)(4) Let A and B be two distinct points and let k be a fixed constant. Then the locus of a point P such that $PA^2 - PB^2 = k$ is a straight line perpendicular to AB .
 (5)(5) In particular, when $k = 0$, it becomes the perpendicular bisector of AB .
 (5)(4) So $AB \perp CD$ iff $AC^2 - AD^2 = BC^2 - BD^2$.

269. (4)(2) (Carnot's theorem(卡諾定理)) Let $\triangle ABC$ be a given triangle and let X, Y, Z be any points.

Then the following conditions are equivalent:

- (i) $\perp(X, BC) \cap \perp(Y, CA) \cap \perp(Z, AB)$;
- (ii) $BX^2 - XC^2 + CY^2 - YA^2 + AZ^2 - ZB^2 = 0$;
- (iii) $\perp(A, YZ) \cap \perp(B, ZX) \cap \perp(C, XY)$.

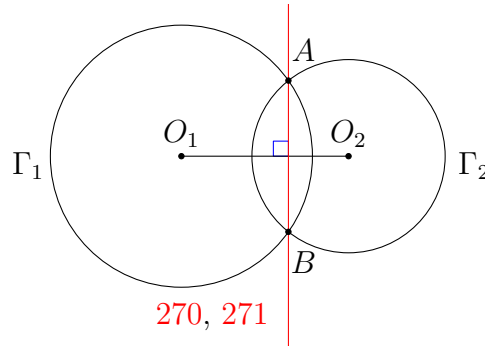


270. (5)(5)(Def) Let Γ_1, Γ_2 be two circles with distinct centres.

Then the locus of a point P such that $\tilde{P}_{\Gamma_1} = \tilde{P}_{\Gamma_2}$ is a straight line, called the *radical axis*(根軸) of Γ_1 and Γ_2 .

(5)(5) In particular, when $\Gamma_1 \cap \Gamma_2 = A, B$, then the radical axis of Γ_1 and Γ_2 is the line AB .

(5)(5) Limiting case: when Γ_1, Γ_2 are tangent at A , then the radical axis of Γ_1 and Γ_2 is $\tan(A, \Gamma_1)$.



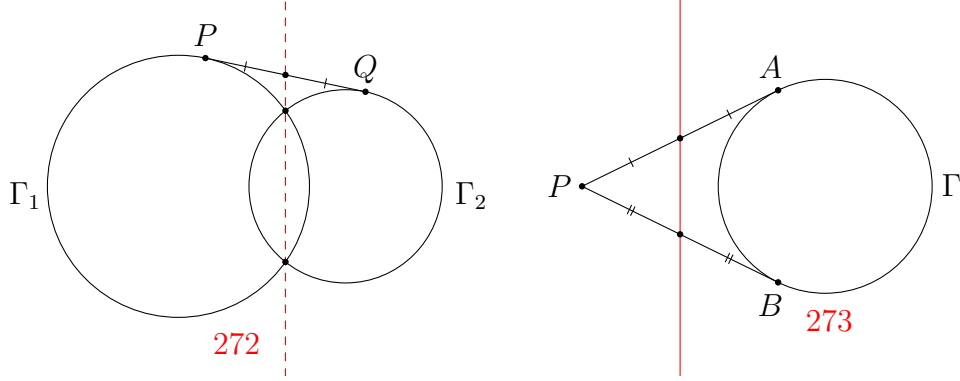
271. (5)(5) Let Γ_1, Γ_2 be two circles with distinct centres O_1, O_2 respectively. Let ℓ be the radical axis of Γ_1 and Γ_2 .

Then $\ell \perp O_1O_2$.

(5)(5) In particular, when $\Gamma_1 \cap \Gamma_2 = A, B$, then $AB \perp O_1O_2$.

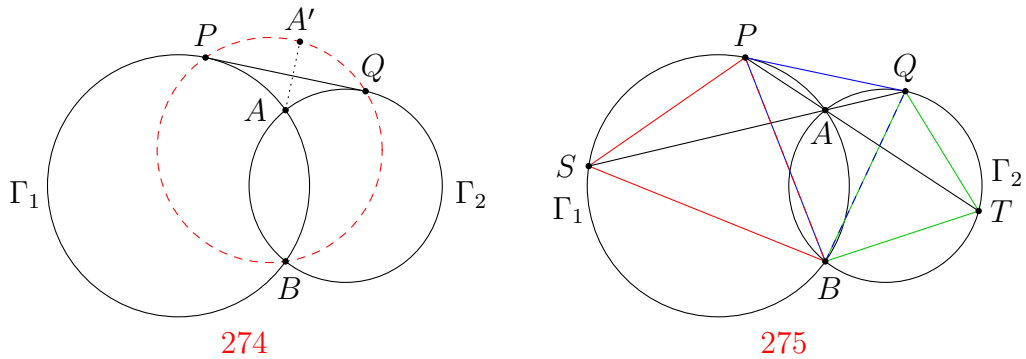
(5)(5) Limiting case: when Γ_1, Γ_2 are tangent at A , then O_1, A, O_2 are collinear.

272. (5)(5) Let Γ_1, Γ_2 be two circles with distinct centres. Let ℓ be an external common tangent of Γ_1 and Γ_2 , and let $P, Q = \ell \cap \Gamma_1, \Gamma_2$. Then M_{PQ} lies on the radical axis of Γ_1 and Γ_2 .



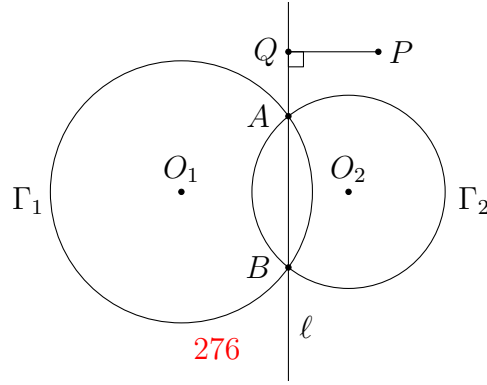
273. (4)(4) Let AB be a chord of a circle Γ which is not a diameter. Let $P = \tan(A, \Gamma) \cap \tan(B, \Gamma)$. Then the radical axis of Γ and the point circle P is the line $M_{PA}M_{PB}$.

274. (2)(1) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let ℓ be an external common tangent of Γ_1 and Γ_2 , and let $P, Q = \ell \cap \Gamma_1, \Gamma_2$. Let $A' = \text{Rf}(A, \ell)$. Then P, B, Q, A' are concyclic ($\sim$).



275. (2)(1) Let Γ_1, Γ_2 be two circles which intersect at A, B . Let ℓ be an external common tangent of Γ_1 and Γ_2 , and let $P, Q = \ell \cap \Gamma_1, \Gamma_2$. Let $S = AQ \cap \Gamma_1$, $T = AP \cap \Gamma_2$. Then $\triangle BSP \sim \triangle BPQ \sim \triangle BQT$ ($\sim$).

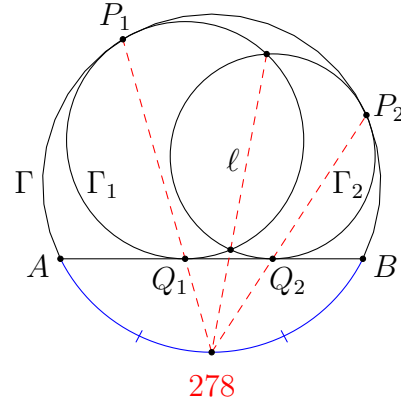
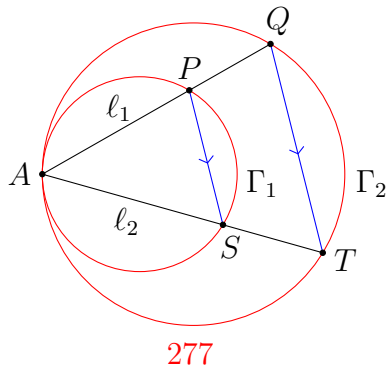
276. (3)(2) Let Γ_1, Γ_2 be two circles with distinct centres O_1, O_2 respectively. Let ℓ be the radical axis of Γ_1 and Γ_2 . Let P be any point, and let $Q = {}_{\perp} P \cap \ell$. Then $\tilde{P}_{\Gamma_1} - \tilde{P}_{\Gamma_2}$ is a linear function in P .
 (1)(1) In particular, $|\tilde{P}_{\Gamma_1} - \tilde{P}_{\Gamma_2}| = 2O_1O_2 \cdot PQ$.



277. (5)(3) Let Γ_1, Γ_2 be two circles which intersect at A . Let $P, Q = \ell_1(A) \cap \Gamma_1, \Gamma_2$, $S, T = \ell_2(A) \cap \Gamma_1, \Gamma_2$.

Then Γ_1 and Γ_2 are tangent at A iff $PS \parallel QT$.

(5)(3) Limiting case: when $\ell_1 = \ell_2$, then Γ_1 and Γ_2 are tangent at A iff $\tan(P, \Gamma_1) \parallel \tan(Q, \Gamma_2)$.



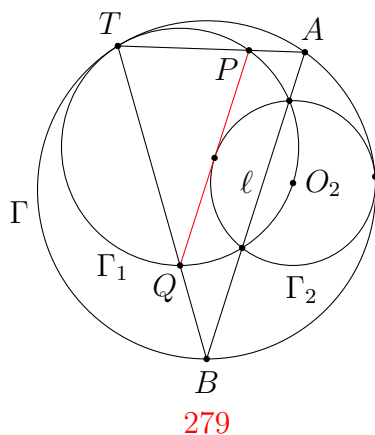
278. (2)(1) Let AB be a chord of a circle Γ . Let Γ_1, Γ_2 be two smaller circles lying on the same side of AB which are internally tangent to Γ at P_1, P_2 and tangent to AB at Q_1, Q_2 respectively. Let ℓ be the radical axis of Γ_1 and Γ_2 .

Then $P_1Q_1 \cap P_2Q_2 \cap \ell$.

(3)(2) In particular, the concurrent point is the midpoint of $\widehat{AB}$.

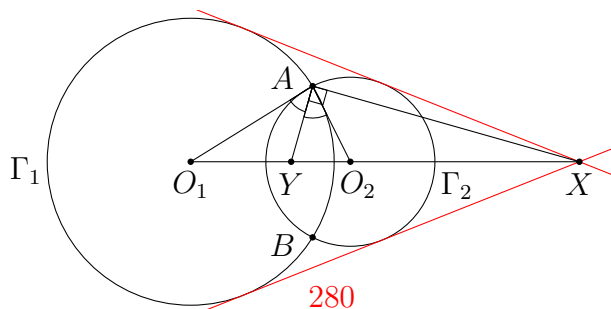
279. (1)(1) In a circle Γ , let Γ_1, Γ_2 be two smaller circles which are internally tangent to Γ such that the centre O_2 of Γ_2 lies on Γ_1 . Let ℓ be the radical axis of Γ_1 and Γ_2 . Let $T = \Gamma \cap \Gamma_1$, $A, B = \ell \cap \Gamma$, $P, Q = TA, TB \cap \Gamma_1$.

Then PQ is tangent to Γ_2 .



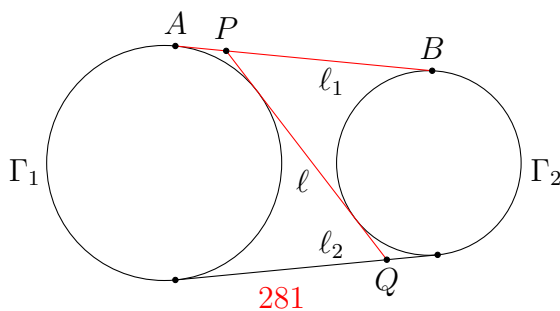
280. (3)(2) Let Γ_1, Γ_2 be two circles with centres O_1, O_2 respectively which intersect at A, B . Let the external and internal angle bisectors of $\angle O_1AO_2$ intersect O_1O_2 at X and Y respectively.

Then the centre of external homothety (i.e. with positive ratio) of Γ_1 and Γ_2 is X , and the centre of internal homothety (i.e. with negative ratio) of Γ_1 and Γ_2 is Y .



281. (2)(1) Let Γ_1, Γ_2 be two circles which do not intersect. Let ℓ be an internal common tangent of Γ_1 and Γ_2 , and let ℓ_1, ℓ_2 be the external common tangents of Γ_1 and Γ_2 . Let $A, B = \ell_1 \cap \Gamma_1, \Gamma_2$ and $P, Q = \ell \cap \ell_1, \ell_2$.

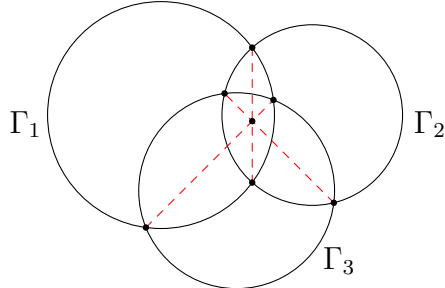
Then $AB = PQ$.



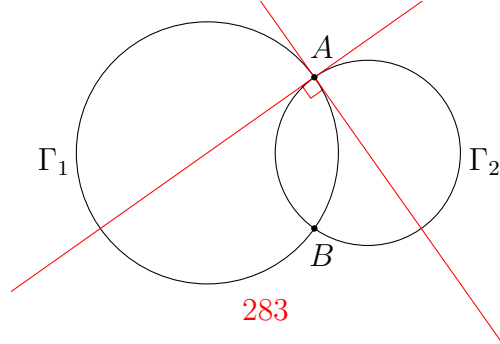
282. (5)(4) Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles with distinct centres. Let ℓ_1 be the radical axis of Γ_2 and Γ_3 , let ℓ_2 be the radical axis of Γ_1 and Γ_3 , and let ℓ_3 be the radical axis of Γ_1 and Γ_2 .

Then $\ell_1 // \ell_2 // \ell_3$ or $\ell_1 \cap \ell_2 \cap \ell_3$.

- (5)(4)(Def) In particular, the intersection point is called the *radical centre* (根心) of $\Gamma_1, \Gamma_2, \Gamma_3$.



282



283

283. (4)(2)(Def) Let Γ_1, Γ_2 be two circles which intersect at A, B .

Then Γ_1 and Γ_2 are called *orthogonal* (正交) if $\tan(A, \Gamma_1) \perp \tan(A, \Gamma_2)$.

284. (2)(1) Let $\Gamma_1, \Gamma_2, \Gamma_3$ be three circles with distinct centres such that their radical centre P lies outside the three circles.

Then the only circle orthogonal to $\Gamma_1, \Gamma_2, \Gamma_3$ is $\odot(P, \sqrt{\widetilde{P}\Gamma_1})$.

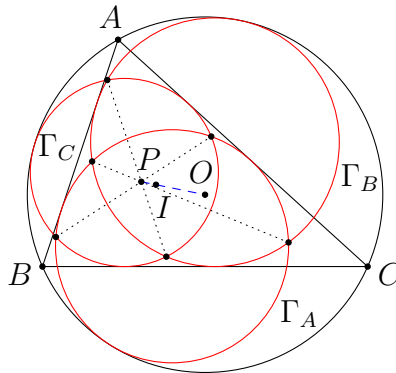
- (2)(1) If P lies inside any of the three circles, then there is no such Γ .

285. (3)(1) Let $\triangle ABC$ be a given triangle.

Then the radical centre of $\odot E_A, \odot E_B, \odot E_C$ is Sp .

286. (3)(1) In $\triangle ABC$, let P be the radical centre of $\Gamma_A, \Gamma_B, \Gamma_C$.

Then O, I, P are collinear.



286

287. (1)(1) (Fuhrmann's theorem) Let A, B, C, D, E, F be any points on a circle Γ in that order. Let $AB = a$, $BC = b$, $CD = c$, $DE = d$, $EF = e$, $FA = f$, $CF = x$, $AD = y$, $BE = z$.

Then $xyz = adx + bey + cfz + ace + bdf$.

Links

Terminologies

- altitude
- anti-Steiner point
- Apollonius circles $\omega_A, \omega_B, \omega_C$
- bicentric quadrilateral
- Brocard angle ω
- Brocard points Br_1, Br_2
- centroid G
- cevian
- cevian triangle
- circumcentre O
- circumcircle $\odot O$
- circumscribed quadrilateral
- concyclic
- cyclic quadrilateral
- diameter
- dumpty points Du_A, Du_B, Du_C
- Euler line
- excentres E_A, E_B, E_C
- excircles $\odot E_A, \odot E_B, \odot E_C$
- extouch triangle
- Fermat points F_1, F_2
- Feuerbach point Fu
- Gergonne point Ge
- harmonic quadrilateral
- humpty points Hu_A, Hu_B, Hu_C
- incentre I
- incentric triangle
- incircle $\odot I$
- inscribed quadrilateral

- internal angle bisector
- intouch triangle
- isodynamic points S_1, S_2
- isogonal
- isogonal conjugate
- isogonal line
- isosceles trapezoid
- kite
- Lemoine point K
- M'Cay cubic
- median
- median triangle
- Miquel point (triangle)
- Miquel point Mq (quadrilateral)
- mixtilinear incircles $\Gamma_A, \Gamma_B, \Gamma_C$
- Nagel line
- Nagel point Na
- Napoleon triangles
- Newton-Gauss line
- nine-point centre N
- nine-point circle $\odot N$
- orthic triangle
- orthocentre H
- orthogonal circles
- parallelogram
- pedal triangle
- power $\tilde{P}_\Gamma$
- radical axis
- radical centre
- rectangle
- rhombus
- Schiffler point Sc

- Sharky-Devil points Sd_A, Sd_B, Sd_C
- Simson line
- Spieker centre Sp
- Spieker circle
- square
- Steiner line
- symmedian
- symmedian point K
- trapezoid

Record

v2.0: adding notations 1–47 and facts 1–196

v2.2: renaming notations 11–17 (12–18 in v2.0), facts 14–21 (13–20 in v2.0), 23–36 (21–34 in v2.0), 38–58 (35–55 in v2.0), 60–69 (56–65 in v2.0), 71–77 (66–72 in v2.0), 80–96 (73–89 in v2.0), 100–107 (90–97 in v2.0), 108–146 (100–138 in v2.0), 148–149 (139–140 in v2.0), 150 (141–142 in v2.0), 151–165 (143–157 in v2.0), 166–184 (159–177 in v2.0) and 185–198 (179–192 in v2.0)
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v3.0: changing to LaTeX

renaming notations 25 (32 in v2.2), 26–32 (25–31 in v2.2), facts 27 (28 in v2.2), 28 (85 in v2.2), 29 (31 in v2.2), 30 (32 in v2.2), 31–34 (74–77 in v2.2), 36–37 (78–79 in v2.2), 38 (29 in v2.2), 39 (82 in v2.2), 40 (39 in v2.2), 41–47 (33–39 in v2.2), 50–72 (40–62 in v2.2), 73 (27 in v2.2), 74 (80 in v2.2), 75–85 (63–73 in v2.2), 86–192 (83–189 in v2.2), 194–202 (190–198 in v2.2)

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v3.1: re-drawing figures and adding hyperlinks

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v3.1.1: adding hyperlinks to figures

v3.1.2: re-drawing figure 40

renaming facts 85–229 (84–228 in v3.1.1)

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v3.1.3: nil

v3.2: re-drawing figures and adding indicators

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